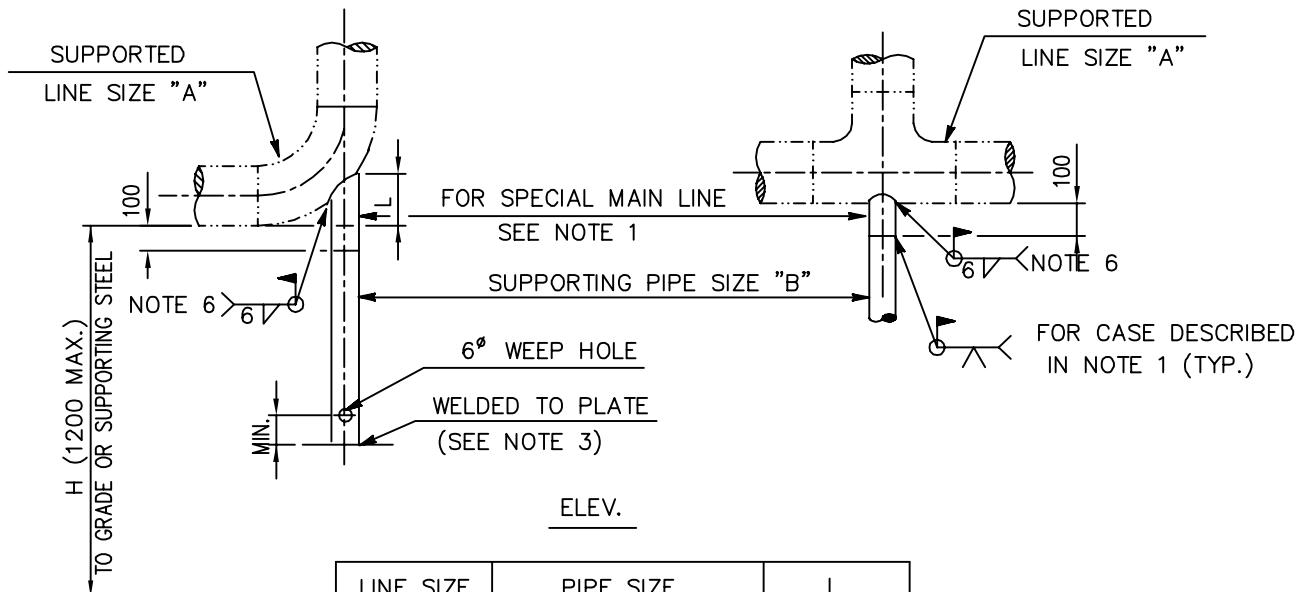


FOR LINE SIZE 2"-24"



LINE SIZE "A"	PIPE SIZE "B"	L (NOTE 5)
2"	1 1/2" SCH.80	71
3"	2" SCH.40	93
4"	3" SCH.40	139
6"	4" SCH.40	186
8"	6" SCH.40	271
10"	8" SCH.40	353
12"	8" SCH.40	370
14"	10" SCH.40	473
16"	10" SCH.40	491
18"	12" STD.WT	572
20"	12" STD.WT	594
24"	12" STD.WT	647

NOTES :

- FOR ALLOY STEEL, STAINLESS STEEL, & STRESS RELIEF LINES, THE MATERIALS SHALL BE THE SAME AS MAIN LINES, AND IT SHALL BE FABRICATED TOGETHER WITH MAIN LINES IN SHOP.
- DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD .
- THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE.

- DESIGNATION NUMBER DENOTE AS FOLLOWS:

01-2B-05E-A

— DENOTE GUIDE ORIENTATION
(FOR LOWER COMPONENT E,F,K,S)

A: AXIAL STOP

G: GUIDE

F: AXIAL STOP & GUIDE

— DENOTE THE LOWER COMPONENT TYPE SEE M-42

— DENOTE DIMENSION "H" (IN 100mm) (SEE NOTE 2)

— DENOTE SUPPORTED LINE SIZE "A"

— DENOTE TYPE NO.

- DIMENSION "L" IS BASED ON LONG RADIUS ELBOW TO WHICH DUMMY PIPE IS WELDED.
- ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT

TYPE-01

ENGINEERING STANDARD

D7TS-701-E

D-1

REV.
1

BY		CHK.	
DATE		APPR.	



PIPE SIZE "A"	L (MAX.)	H (MAX.)	K	M	B	MEMBER "N"
3" SCH. 40	500	1500	70	160	110	C100X50X5
3" SCH. 40	500	2500 (NOTE 4)	70	160	110	C100X50X5
4" SCH. 40	800	2500	85	185	135	C125X65X6

1. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD.

2. THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL, USE TYPE-G, R & T ONLY.

3. DESIGNATION NUMBER DENOTE AS FOLLOWS:

08-3B-0515G

DENOTE THE LOWER COMPONENT TYPE (SEE M-42)

DENOTE DIMENSION "H" (IN 100mm)

DENOTE DIMENSION "L" (IN 100 mm)

DENOTE SUPPORTING PIPE SIZE "A"

DENOTE TYPE NO.

4. ONLY FOR 2" OR SMALLER SINGLE LINE, IF $1500 < H < 2500$.

5. ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT

TYPE-08

ENGINEERING STANDARD

D7TS-701-E

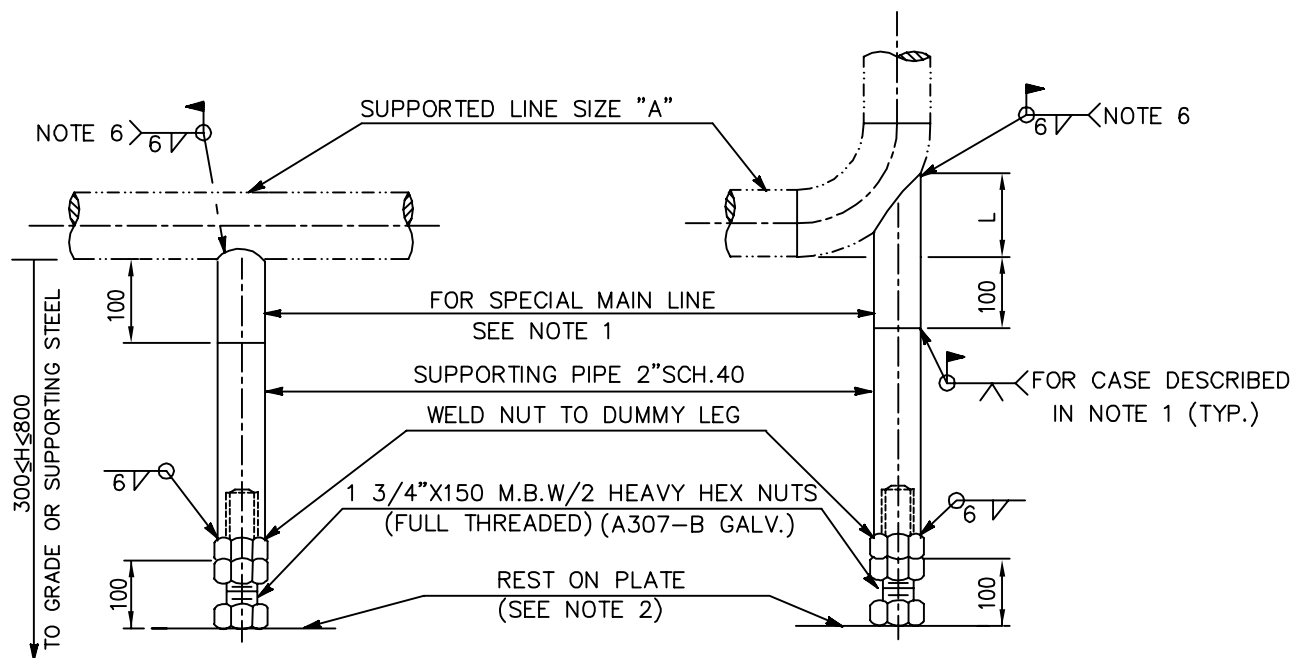
D-8

REV.
1

BY
DATE

CHK.
APPR

FOR LINE SIZE 2"-4"

ELEV.

LINE SIZE "A"	L (NOTE 5)
2"	106
3"	93
4"	106

NOTES :

1. FOR ALLOY STEEL, STAINLESS STEEL & STRESS RELIEF LINES, THE MATERIALS SHALL BE THE SAME AS MAIN LINES, AND IT SHALL BE FABRICATED TOGETHER WITH MAIN LINE IN SHOP.
2. THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL, USE TYPE-B, C H&R ONLY, AND NO NEED BASE PLATE UNDER MACHINE BOLT.

3. DESIGNATION NUMBER DENOTE AS FOLLOWS :

09-2B-05B

- └─┬─┐ DENOTE THE LOWER COMPONENT TYPE SEE M-42
- └─┬─┐ DENOTE DIMENSION "H" (IN 100mm)
- └─┬─┐ DENOTE SUPPORTED LINE SIZE "A"
- └─┬─┐ DENOTE TYPE NO.

4. DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD .
5. DIMENSION "L" IS BASED ON LONG RADIUS ELBOW TO WHICH DUMMY PIPE IS WELDED.
6. ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT
TYPE-09

ENGINEERING STANDARD

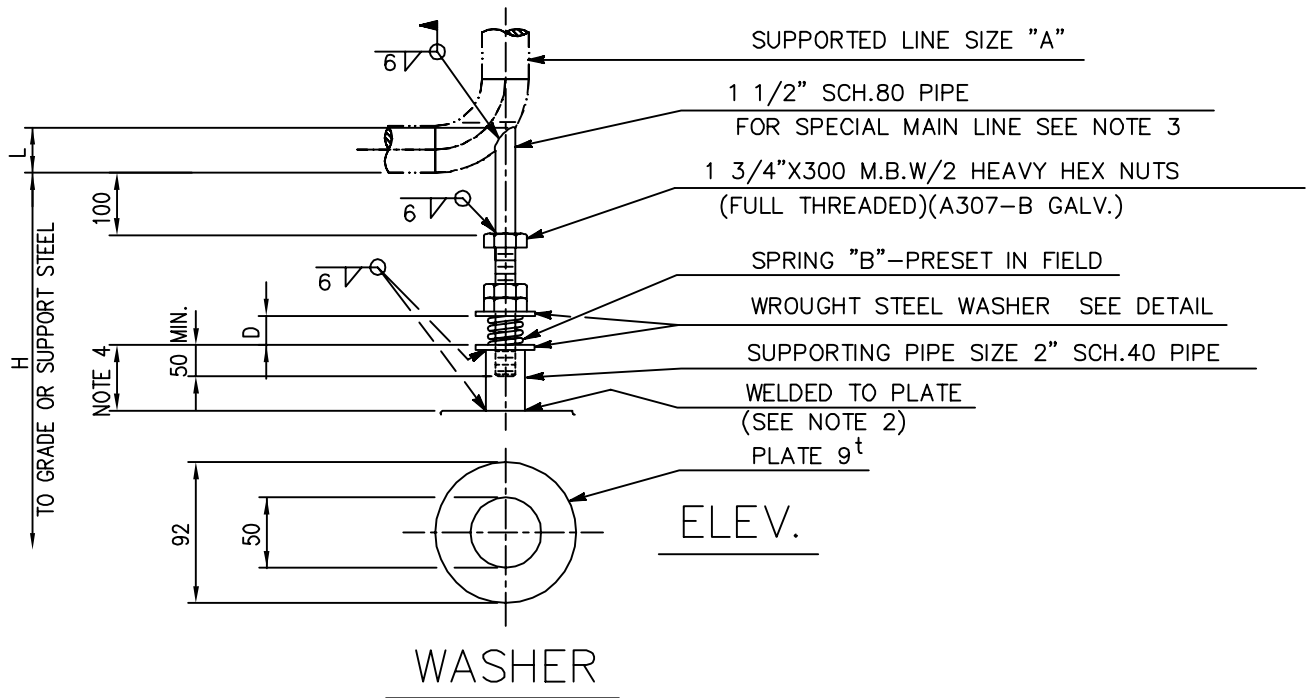
D7TS-701-E

D-9

REV.
1

BY		CHK.	
DATE		APPR.	

FOR LINE SIZE 2"-4"



LINE SIZE "A"	H	SPRING "B" (MAT'L : ASTM A229)	SPRING MARK NO.
2"	600	12 ϕ WIRE , 46 mm I.D.	SPR12
2"	1200	4 ACTIVE COILS, 2 INACTIVE COILS	
3"	600	SPRING CONSTANT = 25 kg/mm	
3"	1200	FREE LENGTH = 100 mm	
4"	600	MAX. RECOM. DEFLECTION : 22 mm	
4"	1200		

NOTES:

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:

11-2B-06G-88

DENOTE TYPE NO. _____

DENOTE SUPPORTED LINE SIZE "A" _____

DENOTE SPRING INSTALLED LENGTH "D"(mm)

DENOTE THE LOWER COMPONENT TYPE
SEE M-42

DENOTE DIMENSION "H"(IN 100mm)

- THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL , USE TYPE-G, R & T ONLY.
- FOR ALLOY STEEL, STAINLESS STEEL & STRESS RELIEF LINES, THE MATERIALS SHALL BE THE SAME AS MAIN LINES.
- DIMENSION SHALL BE CUT TO SUIT IN FIELD.
- WHEN THIS TYPE OF SUPPORT IS APPLIED ADJACENT TO PUMP OR TURBINE CONNECTION, THE SPRING MUST BE ADJUSTED BY TRIAL TO PERMIT EASY REMOVE OF ALL FLANGE BOLTS, AND IT IS NOT PERMISSIBLE TO CORRECT PIPING MISALIGNMENT BY COMPRESSING THE SPRING TO ITS SOLID LENGTH.
- DIMENSION "L" IS BASED ON LONG RADIUS ELBOW TO WHICH DUMMY PIPE IS WELDED.
- ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT

TYPE-11

ENGINEERING STANDARD

D7TS-701-E

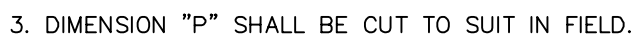
D-11

REV.
1BY
DATECHK.
APPR.



STANDARD DIMENSION LIST												
A	D	F	K	M	P	Q	B	I	J	T	MEMBER "N"	H (MAX.)
4" SCH.40	260	12	150	160	120	170	135	84	47	6	H100X100X6X8	3000
6" SCH.40	380	16	200	210	180	190	190	130	72	6	H150X150X7X10	3000
8" SCH.40	380	16	200	210	150	210	240	130	72	6	H150X150X7X10	4000
10" SCH.40	450	16	250	260	180	220	295	176	96	9	H200X200X8X12	4000
12" STD.WT	500	16	250	260	190	240	345	176	96	9	H200X200X8X12	4000

15-4B-1030



TYPE-15

D7TS-701-E

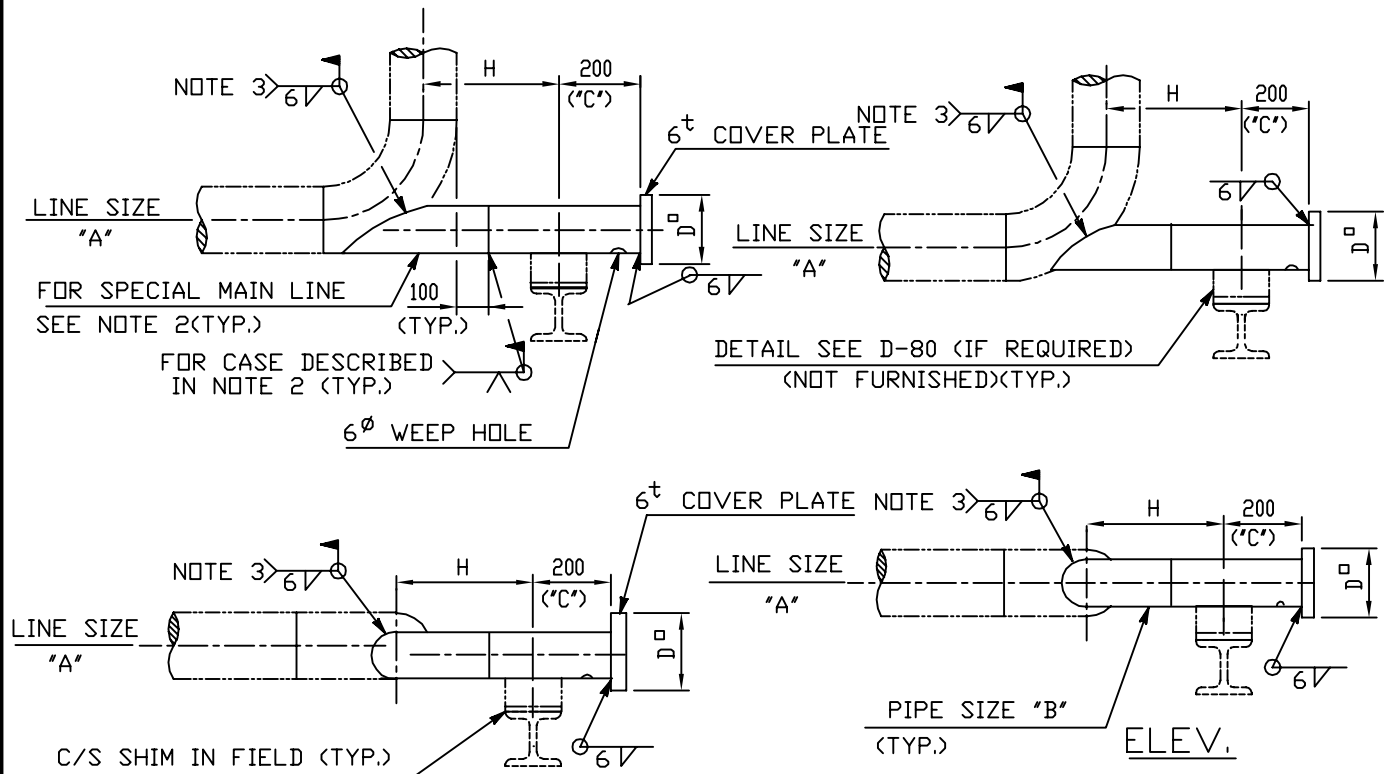
D-16

REV.
1

BY
DATE

CHK.
APPR

FOR LINE SIZE 2"-24"



LINE SIZE "A"	PIPE SIZE "B"	D (mm)
2"	1 1/2" SCH.80	70
3"	2" SCH.40	80
4"	3" SCH.40	110
6"	4" SCH.40	140
8"	6" SCH.40	190
10"	8" SCH.40	240
12"	10" SCH.40	290
14"	12" STD.WT	340
16"	12" STD.WT	340
18"	14" STD.WT	380
20"	14" STD.WT	380
24"	16" STD.WT	430

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

16-2B-04-03

MODIFY DIMENSION "C" (IN 100mm) IF ANY
 DENOTE DIMENSION "H" (IN 100mm)
 DENOTE LINE SIZE "A"
 DENOTE TYPE NO.

2. FOR ALLOY STEEL, STAINLESS STEEL & STRESS RELIEF LINES, THE MATERIALS SHALL BE THE SAME AS MAIN LINES, AND IT SHALL BE FABRICATED TOGETHER WITH MAIN LINES IN SHOP.
3. ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT

TYPE-16

ENGINEERING STANDARD

D7TS-701-E

D-18

 REV.
 1

BY		CHK.	
DATE		APPR.	

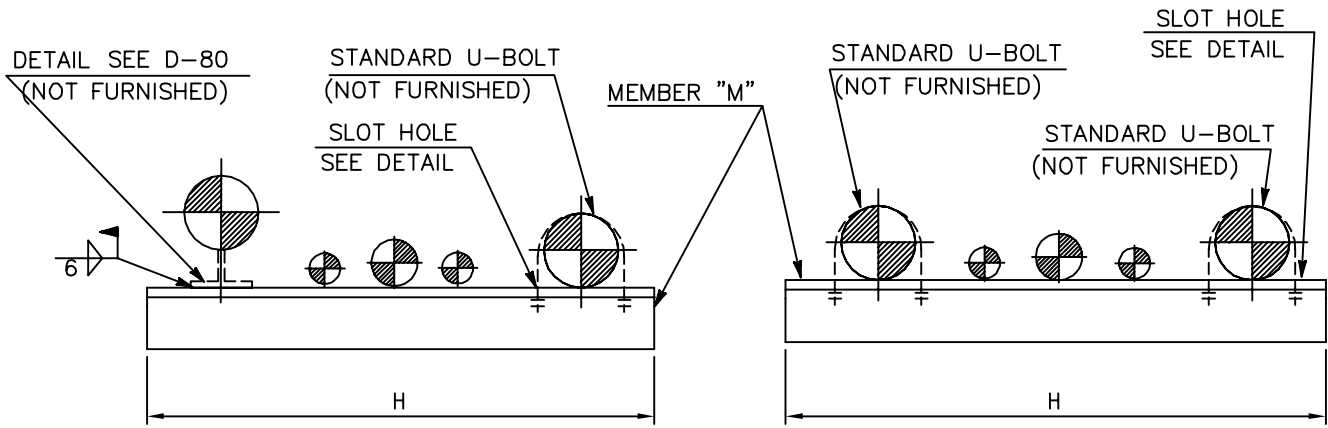
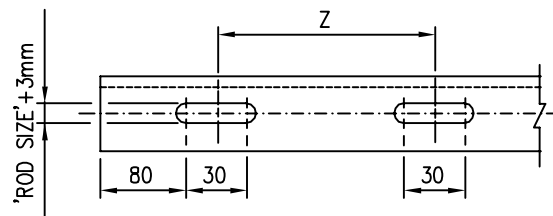


FIG. A

ELEV.

FIG. B



SLOT HOLE DETAIL

LINE SIZE	Z (mm)
2"	76
3"	104
4"	130
6"	184
8"	235
10"	286
12"	340

MEMBER "M"	H (MAX.)
L50X50X6	1000
L75X75X9	1500
C125X65X6X8	2000

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

20 -L50 -05A

DENOTE FIG.-A
 DENOTE LENGTH "H" (IN 100mm)
 DENOTE MEMBER "M"
 DENOTE TYPE NO.

2. DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD .



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DETAIL OF PIPE SUPPORT
 TYPE-20

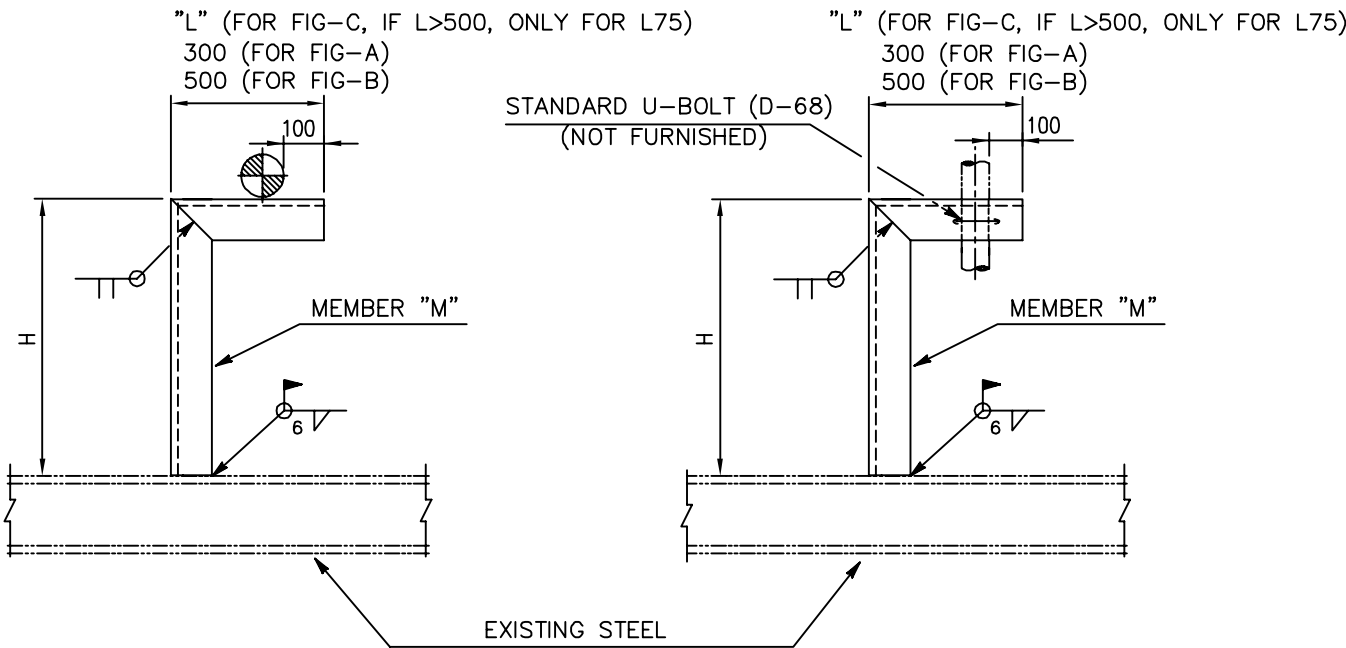
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-22

 REV.
 1

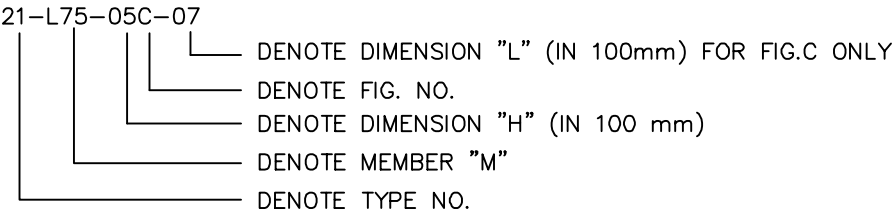


ELEV.

MEMBER "M"	H (MAX.)	L (MAX.)
L50X50X6	1000	500
L75X75X9	1500	800

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :



2. DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD .



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DETAIL OF PIPE SUPPORT
TYPE-21

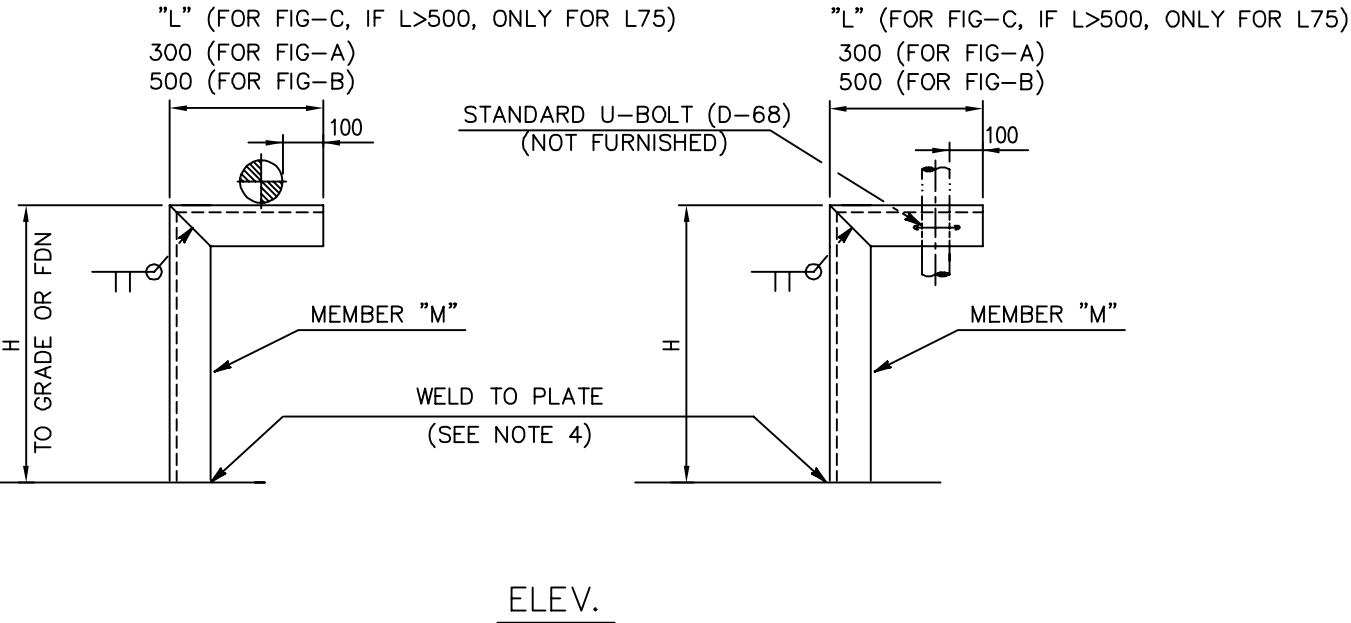
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-23

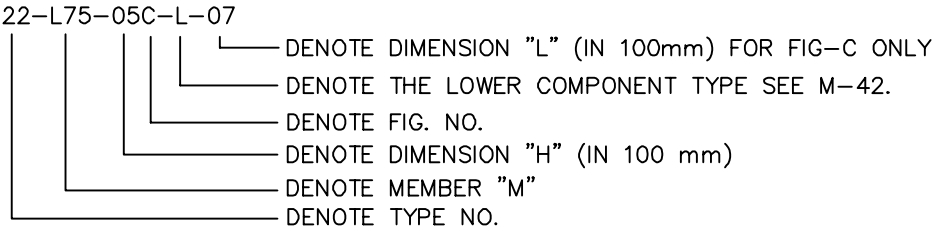
REV.
1



MEMBER "M"	H (MAX.)	L (MAX.)
L50X50X6	1000	500
L75X75X9	1500	800

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :



2. DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD .

3. IF THE FOUNDATION IS NOT USED IN THIS DETAIL, DIMENSION "H" SHALL BE COUNTED FROM THE ELEVATION OF THE LOWEST POINT OF PAVING.

4. THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL, USE TYPE L & T ONLY.



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DETAIL OF PIPE SUPPORT
TYPE-22

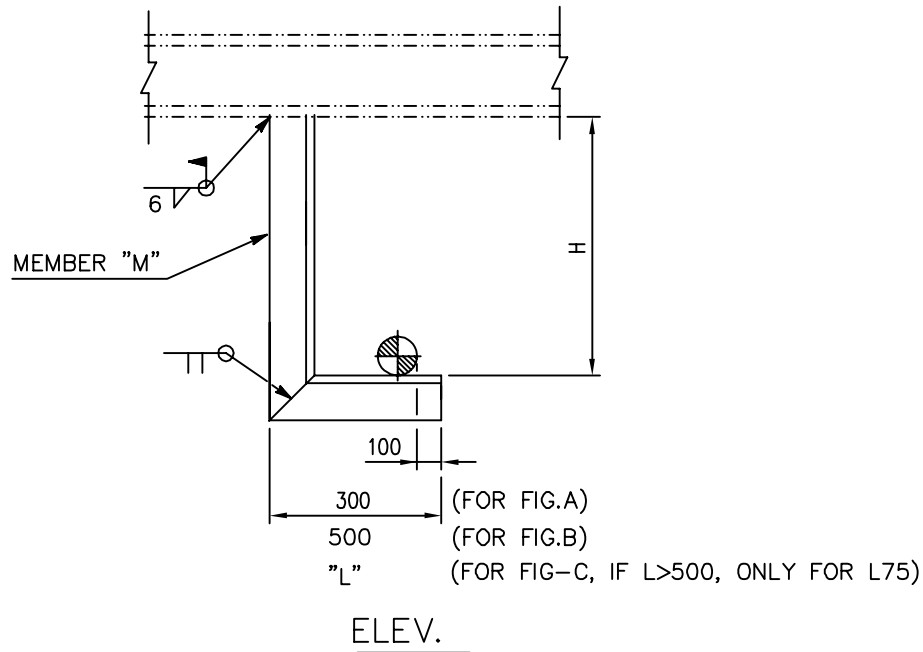
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-24

REV.
1



MEMBER "M"	H (MAX.)	L (MAX.)
L50X50X6	1000	500
L75X75X9	2000	800

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:

23-L75-05C-07

- DENOTE DIMENSION "L" (IN 100mm) FOR FIG.C ONLY
- DENOTE FIG. NO.
- DENOTE DIMENSION "H" (IN 100 mm)
- DENOTE MEMBER "M"
- DENOTE TYPE NO.

2. DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD.



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DETAIL OF PIPE SUPPORT
TYPE-23

ENGINEERING STANDARD

D7TS-701-E

D-25

REV.
1

BY		CHK.	
DATE		APPR.	

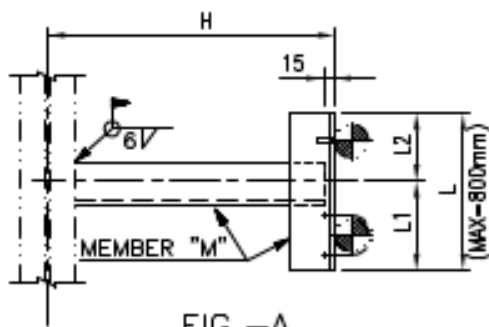


FIG.-A
(PLAN VIEW)

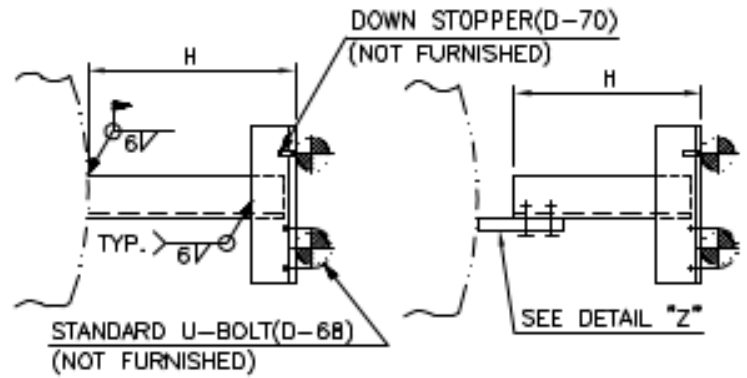


FIG.-B
(PLAN VIEW)

FIG.-C
(PLAN VIEW)

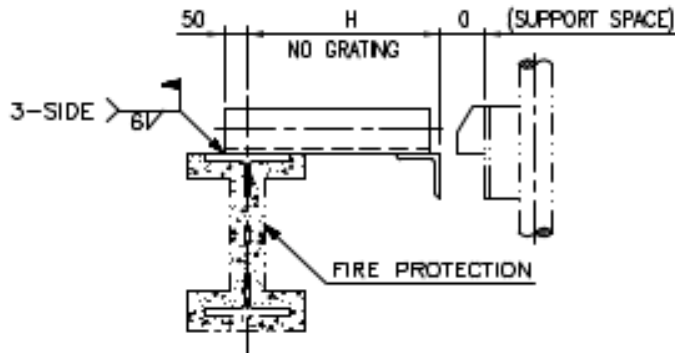
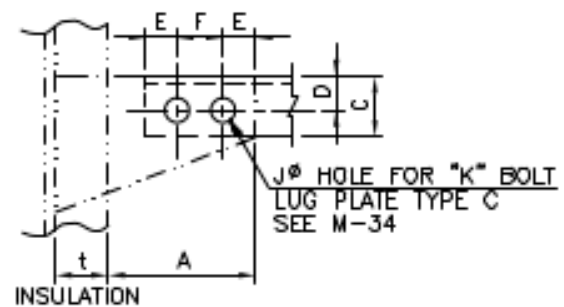


FIG.-A
(ELEV. VIEW FOR W/ F. P.)



DETAIL "Z"

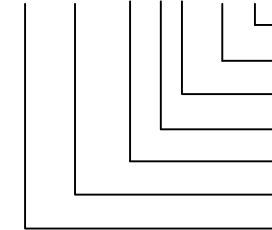
DIMENSIONS (mm)

MEMBER "M"	A	C	D	E	F	JØ	K	L (MAX.)	H (MAX.)
L50X50X6	150	50	30	30	60	19	5/8"X40	500	800
L75X75X9	160	75	40	30	70	22	3/4"X50	800	1200

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

25-L50-0505A-0401



MODIFY DIMENSION "L2" (IN 100mm) IF ANY
MODIFY DIMENSION "L1" (IN 100mm) IF ANY
DENOTE FIG. NO.
DENOTE DIM. "H" (IN 100mm)
DENOTE DIM. "L" (IN 100mm)
DENOTE MEMBER "M"
DENOTE TYPE NO.

2. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD.



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DETAIL OF PIPE SUPPORT
TYPE-25

ENGINEERING STANDARD

D7TS-701-E

D-27

REV.
1

BY		CHK.	
DATE		APPR.	

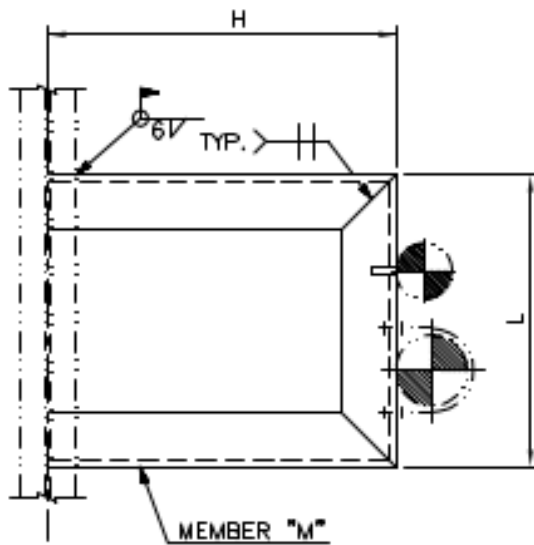


FIG.-A
(PLAN VIEW)

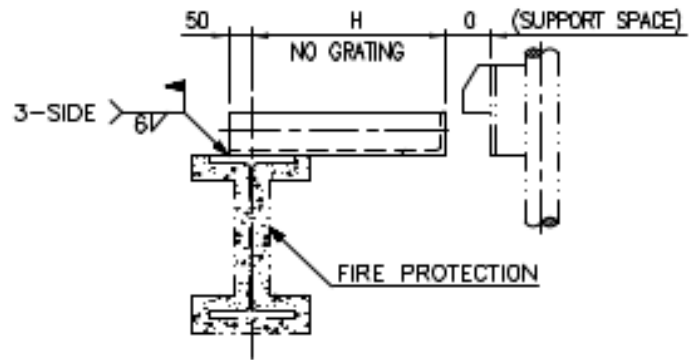


FIG.-A
(ELEV. VIEW FOR W/ F. P.)

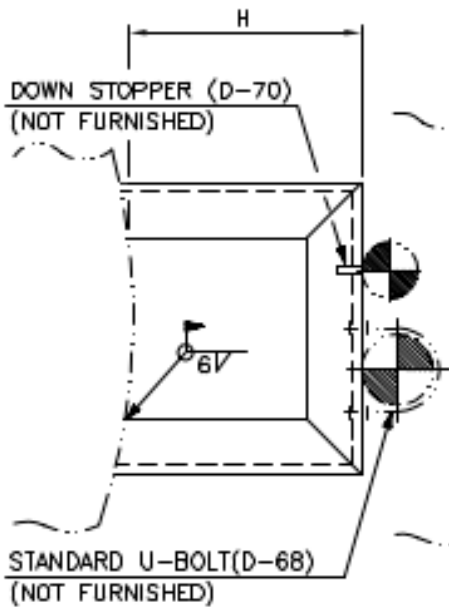


FIG.-B
(PLAN VIEW)

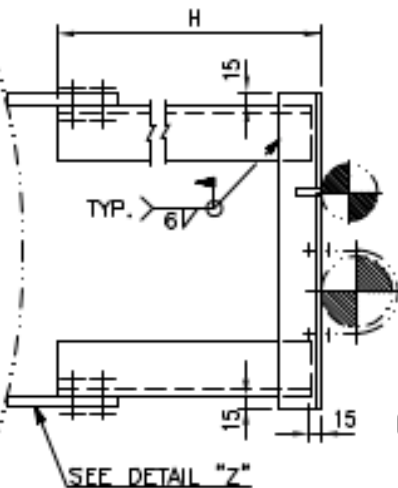
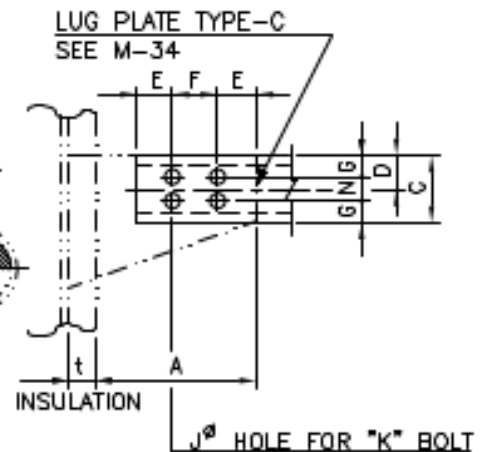


FIG.-C
(PLAN VIEW)



DETAIL "Z"

NOTES:

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:

26-L75-1005A

— DENOTE FIG. NO.
— DENOTE DIM. "H" (IN 100mm)
— DENOTE DIM. "L" (IN 100mm)
— DENOTE MEMBER "M"
— DENOTE TYPE NO.

2. DIMENSIONS "L" & "H" SHALL BE CUT TO SUIT IN FIELD.

3. FOR DIMENSIONAL DATA, SEE SH'T D-29.



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DETAIL OF PIPE SUPPORT

TYPE-26

(1 OF 2)

ENGINEERING STANDARD

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REV.
1

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MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	600	800
L75X75X9	1000	1200
C125X65X6	1500	1500
C150X75X9	1500	1500

DIMENSION (mm)

MEMBER "M"	A	C	D	E	F	G	N	J ϕ	K
L50X50X6	150	50	30	30	60	—	—	19	5/8"X40
L75X75X9	160	75	40	30	70	—	—	22	3/4"X50
C125X65X6	170	125	—	30	80	35	55	22	3/4"X50
C150X75X9	190	150	—	30	100	40	70	22	3/4"X50



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DETAIL OF PIPE SUPPORT

TYPE-26

(2 OF 2)

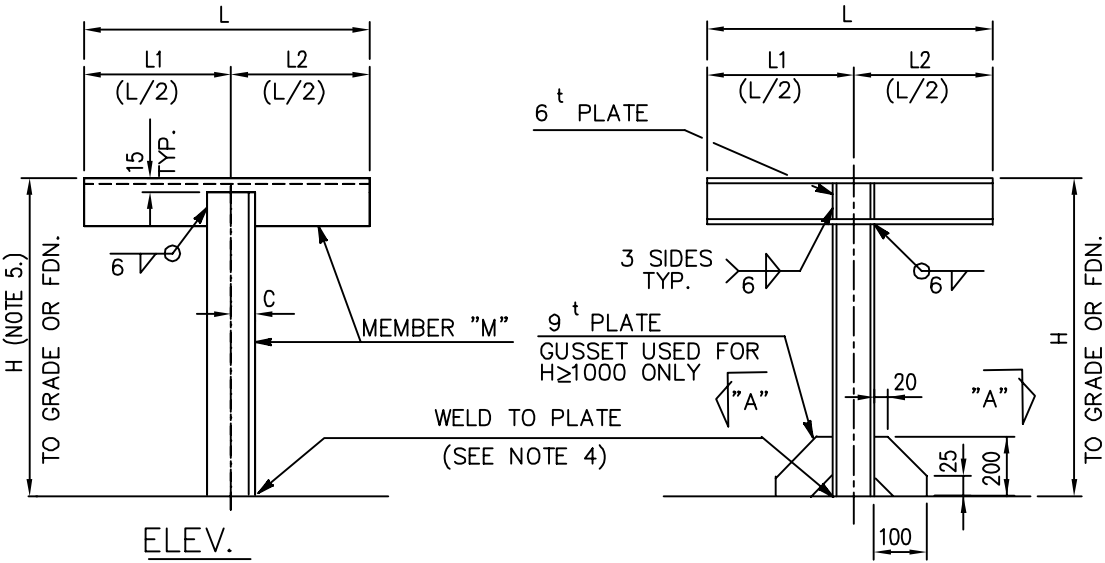
ENGINEERING STANDARD

D7TS-701-E

D-29

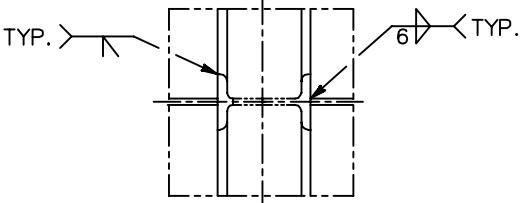
REV.
1

BY			CHK.		
DATE			APPR.		



DIMENSION (mm)

MEMBER "M"	L (MAX.)	H (MAX.)	C
L50X50X6	500	800	30
L75X75X9	800	1300	35
L100X100X10	1000	1500	50
H125X125X6.5X9	1000	1500	—
H150X150X7X10	1000	2500	—
H200X200X8X12	1000	2500	—



NOTES:

1. DESIGNATION DENOTE AS FOLLOWS:

SECTION "A" ~ "A"

27-L75-0505L-0401

- MODIFY DIMENSION "L2" (IN 100mm) IF ANY
- MODIFY DIMENSION "L1" (IN 100mm) IF ANY
- DENOTE THE LOWER COMPONENT TYPE SEE M-42
- DENOTE DIM. "H" (IN 100 mm)
- DENOTE DIM. "L" (IN 100 mm)
- DENOTE MEMBER "M"
- DENOTE TYPE NO.

- 2. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD.
- 3. IF THE FOUNDATION IS NOT USED IN THIS DETAIL, DIMENSION "H" SHALL BE COUNTED FROM THE ELEVATION OF THE LOWEST POINT OF PAVING.
- 4. THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL, USE TYPE-L,R & T ONLY.



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DETAIL OF PIPE SUPPORT
TYPE-27

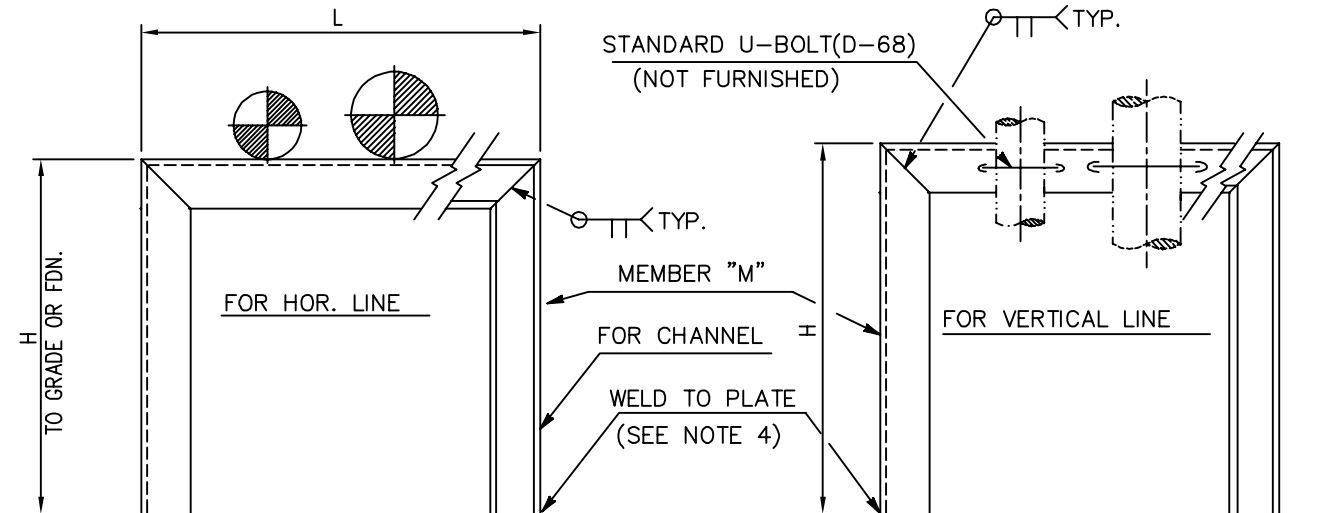
ENGINEERING STANDARD

D7TS-701-E

D-30

REV.
1

BY		CHK.	
DATE		APPR.	



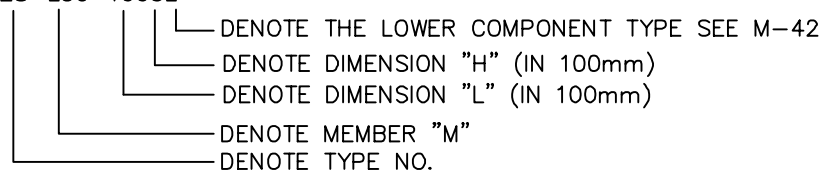
ELEV.

MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	1000	500
L75X75X9	1000	1500
C125X65X6	1500	1500
C150X75X9	1500	2000

NOTES:

1. DESIGNATION NUMBER DENOTES AS FOLLOWS:

28-L50-1005L



2. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD .
3. IF THE FOUNDATION IS NOT USED IN THIS DETAIL, DIMENSION "H" SHALL BE COUNTED FROM THE ELEVATION OF THE LOWEST POINT OF PAVING.
4. THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL, USE TYPE-L & T ONLY.



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DETAIL OF PIPE SUPPORT
TYPE-28

ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-31

REV.
1

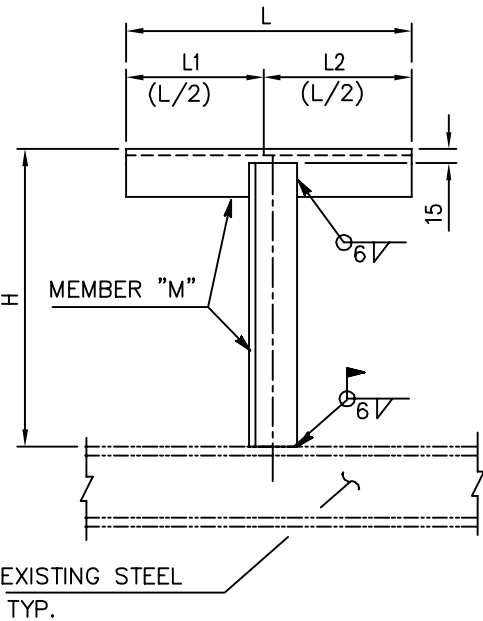


FIG.-A
(ELEV.)

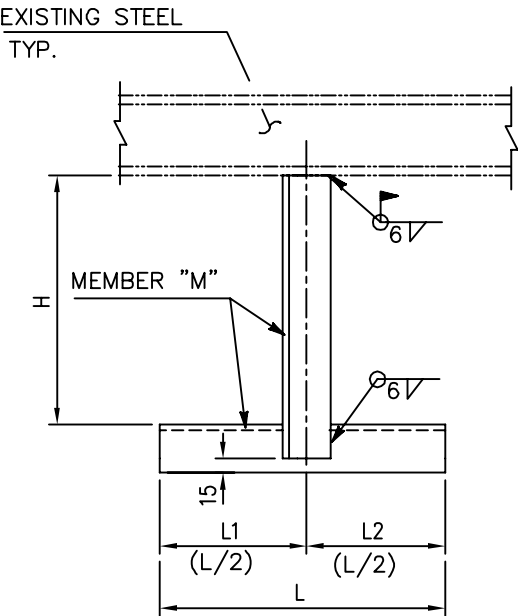
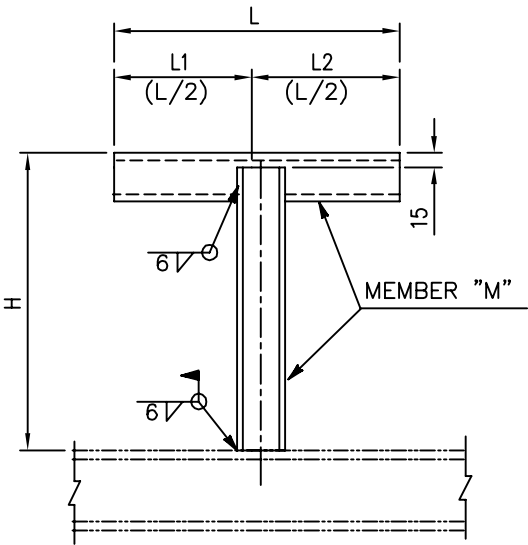
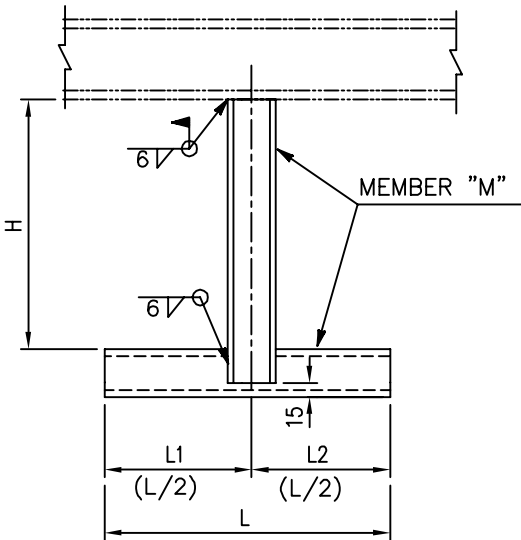
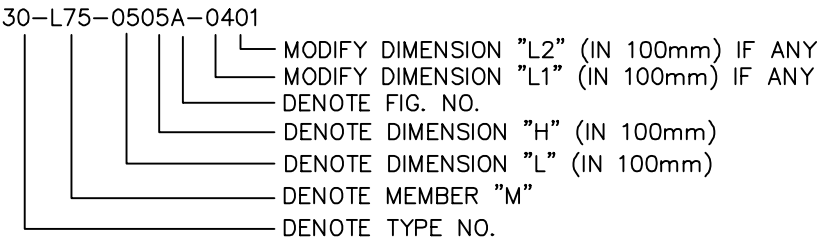


FIG.-B
(ELEV.)



DIMENSIONS (mm)		
MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	500	800
L75X75X9	800	1200
C125X65X6	1000	1500
C150X75X9	1000	2000

NOTES: 1. DESIGNATION NUMBER DENOTE AS FOLLOWS:



2. DIMENSION "H" & "L" SHALL BE CUT SUIT IN FIELD.



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DETAIL OF PIPE SUPPORT
TYPE-30

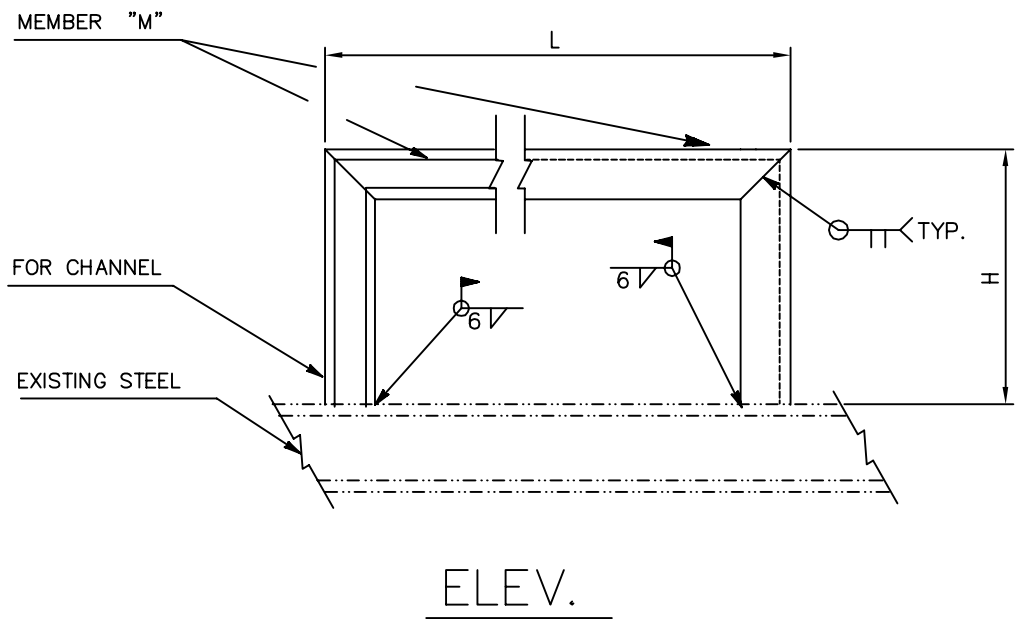
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-35

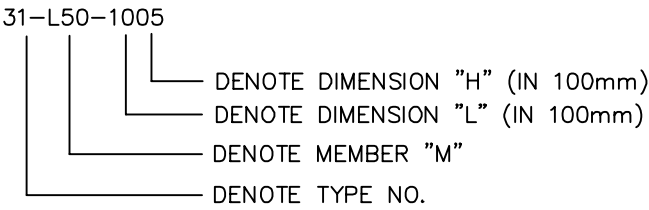
REV.
1



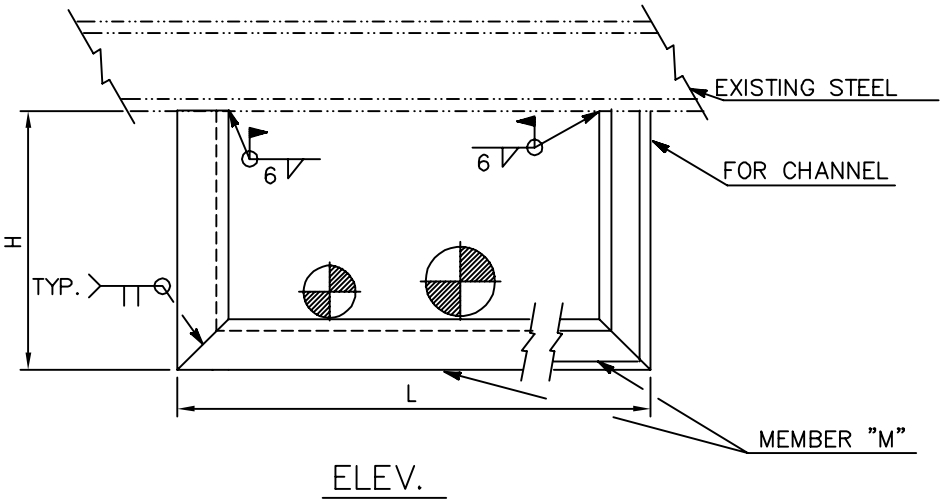
MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	1000	1000
L75X75X9	1000	1500
C125X65X6	1500	1500
C150X75X9	2000	2000

NOTES : 1. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD .

2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

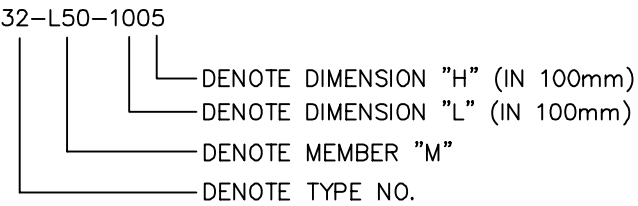


 中鼎工程股份有限公司 CTCI CORPORATION						ENGINEERING STANDARD	
DETAIL OF PIPE SUPPORT TYPE-31						D7TS-701-E	
						D-36	REV. 1
BY			CHK.				
DATE			APPR.				



MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	1000	1000
L75X75X9	1000	1500
C125X65X6	1500	1500
C150X75X9	2000	2000

NOTES : 1. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD .
2. DESIGNATION NUMBER DENOTE AS FOLLOWS :



中鼎工程股份有限公司
CTCI CORPORATION

DETAIL OF PIPE SUPPORT
TYPE-32

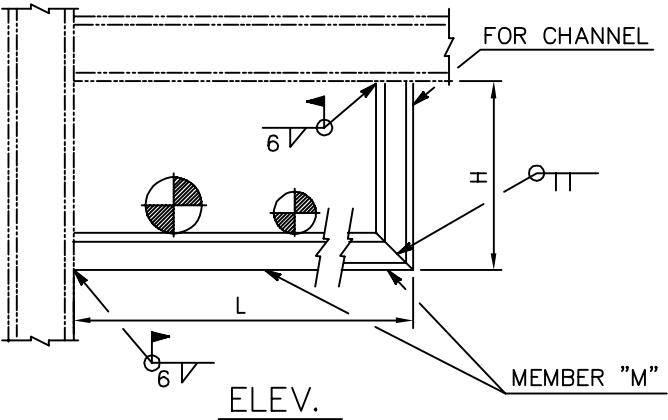
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-37

REV.
1



MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	1000	1000
L75X75X9	1000	1500
C125X65X6	1500	1500
C150X75X9	2000	2000

- NOTES :
- 1. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD.
 - 2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

33-L50-1005

- DENOTE DIMENSION "H" (IN 100mm)
- DENOTE DIMENSION "L" (IN 100mm)
- DENOTE MEMBER "M"
- DENOTE TYPE NO.



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DETAIL OF PIPE SUPPORT
TYPE-33

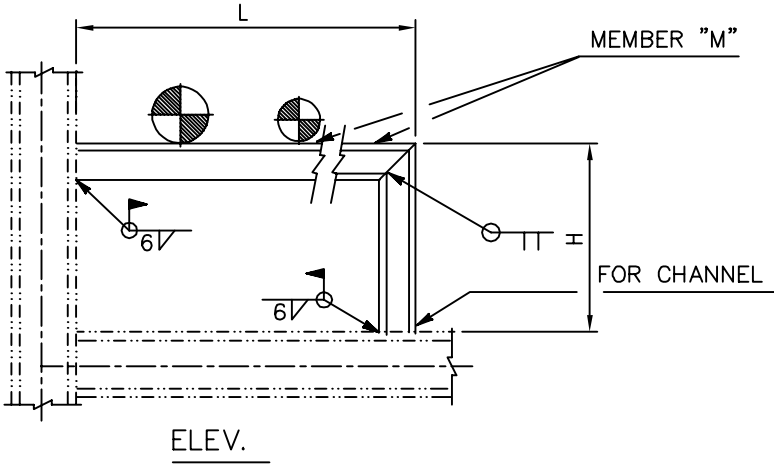
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

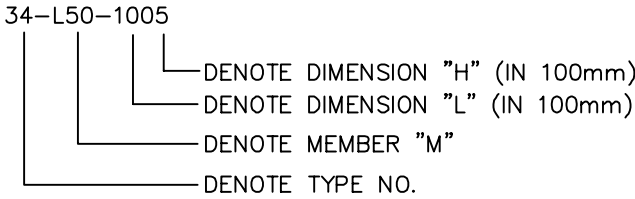
D-38

REV.
1



MEMBER "M"	L (MAX.)	H (MAX.)
L50X50X6	1000	1000
L75X75X9	1000	1500
C125X65X6	1500	1500
C150X75X9	2000	2000

- NOTES : 1. DIMENSION "H" & "L" SHALL BE CUT TO SUIT IN FIELD .
2. DESIGNATION NUMBER DENOTE AS FOLLOWS :



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CTCI CORPORATION

DETAIL OF PIPE SUPPORT
TYPE-34

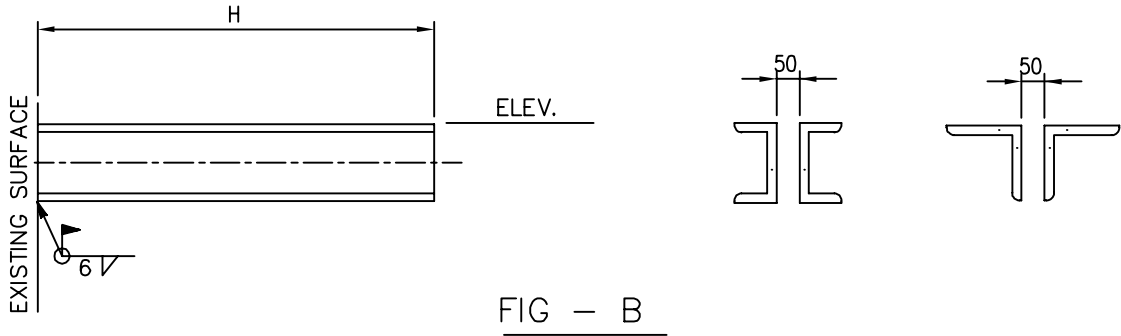
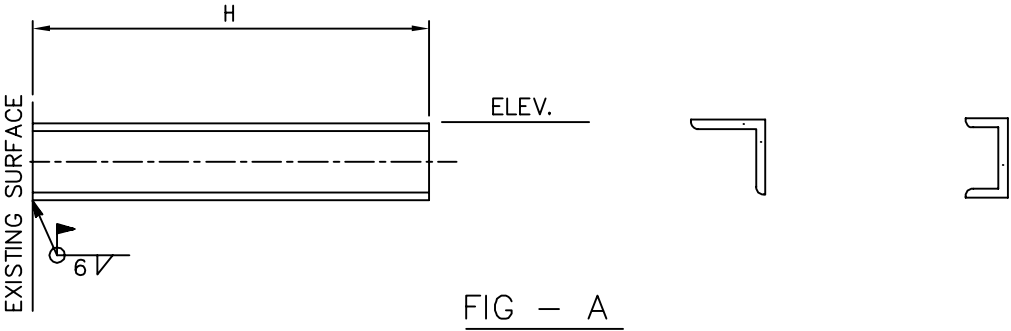
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-39

REV.
1

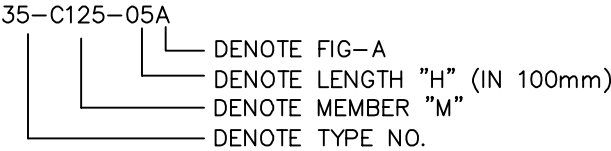


DIMENSIONS (mm)

MEMBER "M"	H (FOR FIG-A) (MAX.)	H (FOR FIG-B) (MAX.)
L50X50X6	600	NOTE 2
L75X75X9	800	900
C100X50X5	1400	900
C125X65X6	1400	900
C150X75X9	1400	NOTE 2

NOTE :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :



2. FOR MEMBER "L50X50X6","C150X75X9" USE FIG-A ONLY.



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DETAIL OF PIPE SUPPORT
TYPE-35

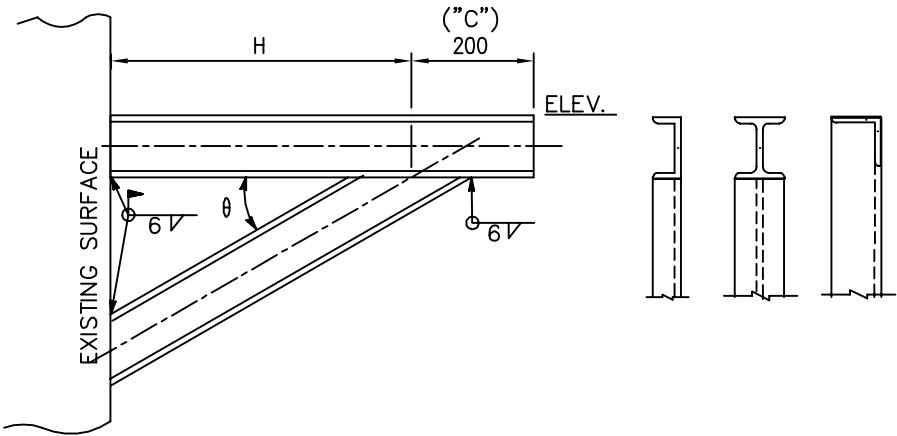
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

D-40

REV.
1



MEMBER "M"	H (MAX.)
L75X75X9	1450
C100X50X5	1450
C125X65X6	1450
H100X100X6	2050
H125X125X6.5	2050
H150X150X7	2050

NOTES:

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :
- 37-C125-1200A OR B-05
- MODIFY DIMENSION "C" (IN 100mm) IF ANY
- SEE NOTE 2
- DENOTE LENGTH "H" (IN mm)
- DENOTE MEMBER "M"
- DENOTE TYPE NO.
2. A: FOR $\theta = 30^\circ$
B: FOR $\theta = 45^\circ$



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DETAIL OF PIPE SUPPORT
TYPE-37

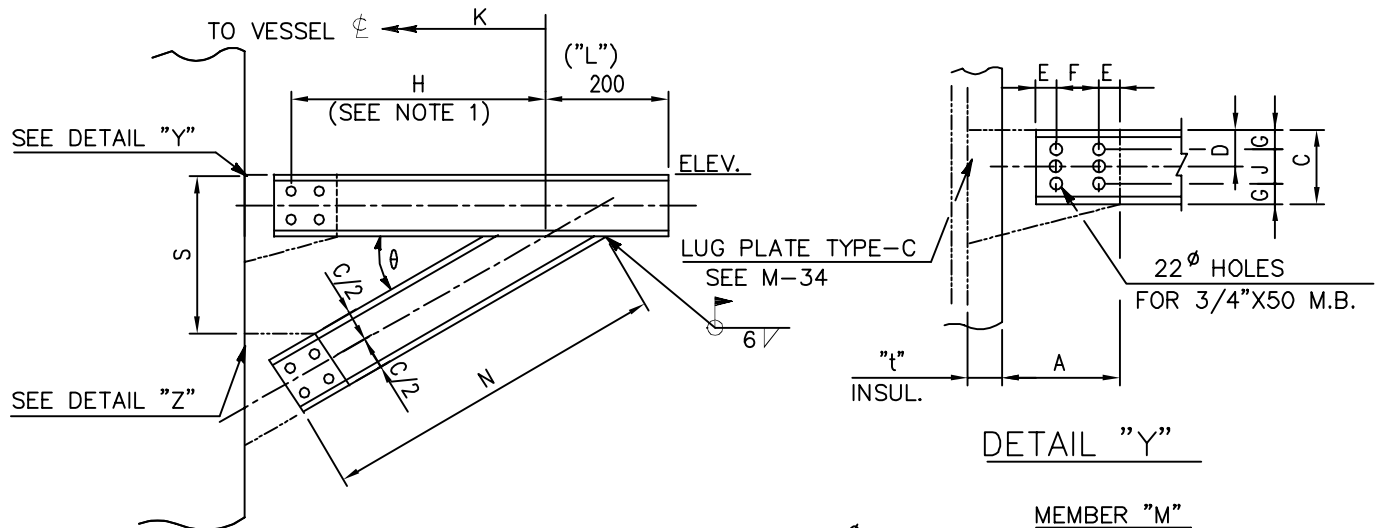
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

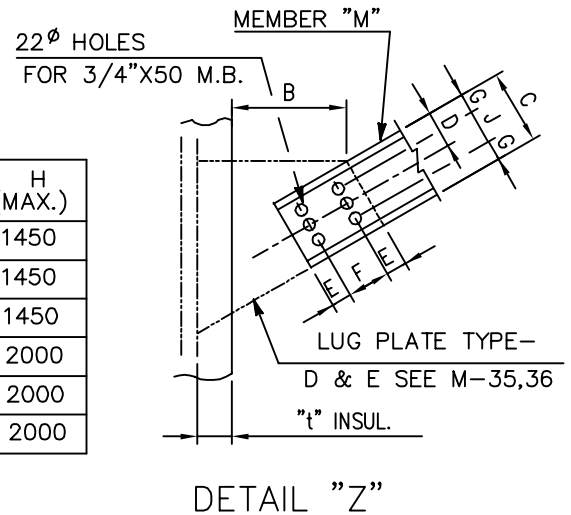
D-42

REV.
1



DIMENSIONS (mm)

MEMBER "M"	A	C	D	E	F	G	J	H (MAX.)
L75X75X9	160	75	40	30	70	—	—	1450
C100X50X5	170	100	50	30	80	—	—	1450
C125X65X6	170	125	—	30	80	35	55	1450
C150X75X9	190	150	—	30	100	40	70	2000
C180X75X7	210	180	—	35	110	45	90	2000
C200X90X8	220	200	—	35	120	50	100	2000



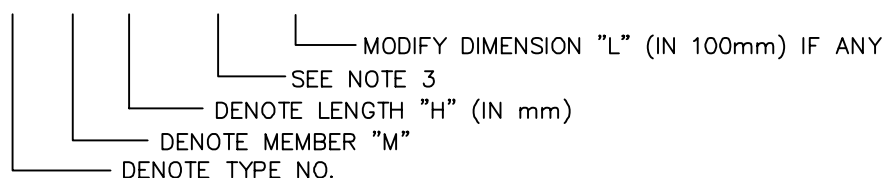
DIMENSIONS (mm)

MEMBER "M"	$\theta = 30^\circ$			$\theta = 45^\circ$		
	B	S	N	B	S	N
L75X75X9	135	0.577H-8	1.155H+ 92	115	H - 30	1.414H+57
C100X50X5	145	0.577H-7	1.155H+100	120	H - 31	1.414H+55
C125X65X6	145	0.577H+4	1.155H+114	130	H - 33	1.414H+41
C150X75X9	165	0.577H+3	1.155H+125	140	H - 36	1.414H+47
C180X75X7	185	0.577H+7	1.155H+145	150	H - 32	1.414H+60
C200X90X8	195	0.577H+9	1.155H+155	160	H - 36	1.414H+56

NOTES :

1. $H=K-R-t-E-30$ WHERE "R" IS RADIUS OF VESSEL
2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

39-C100-500 A OR B-05



3. A: FOR $\theta = 30^\circ$
B: FOR $\theta = 45^\circ$



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DETAIL OF PIPE SUPPORT

TYPE-39

ENGINEERING STANDARD

D7TS-701-E

D-45

REV.
1

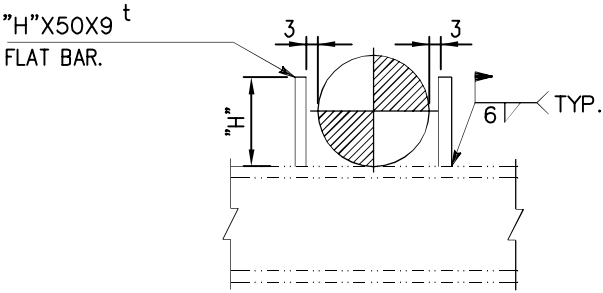
BY

DATE

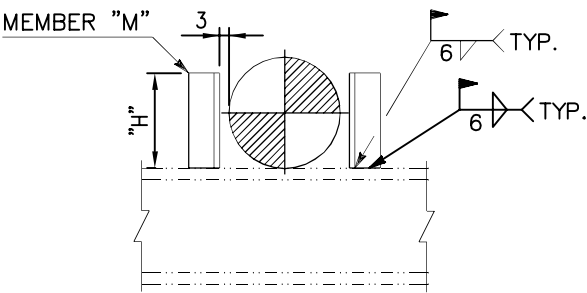
CHK.

APPR.

FOR LINE SIZE 3/4"-24"



(3" & SMALLER)



(4" TO 24")

ELEV.

LINE SIZE	MEMBER "M"	H
3/4"	—	25
1"	—	30
1 1/2"	—	45
2"	—	60
2 1/2"	—	70
3"	—	80
4"	L50X50X6	125
5"	L50X50X6	125
6"	L50X50X6	125
8"	L75X75X9	150
10"	L75X75X9	150
12"	L75X75X9	200
14"	L75X75X9	200
16"	L75X75X9	250
18"	L75X75X9	300
20"	L75X75X9	300
24"	L75X75X9	350

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

51-2B
— DENOTE LINE SIZE
— DENOTE TYPE NO.



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DETAIL OF PIPE SUPPORT
TYPE-51

ENGINEERING STANDARD

D7TS-701-E

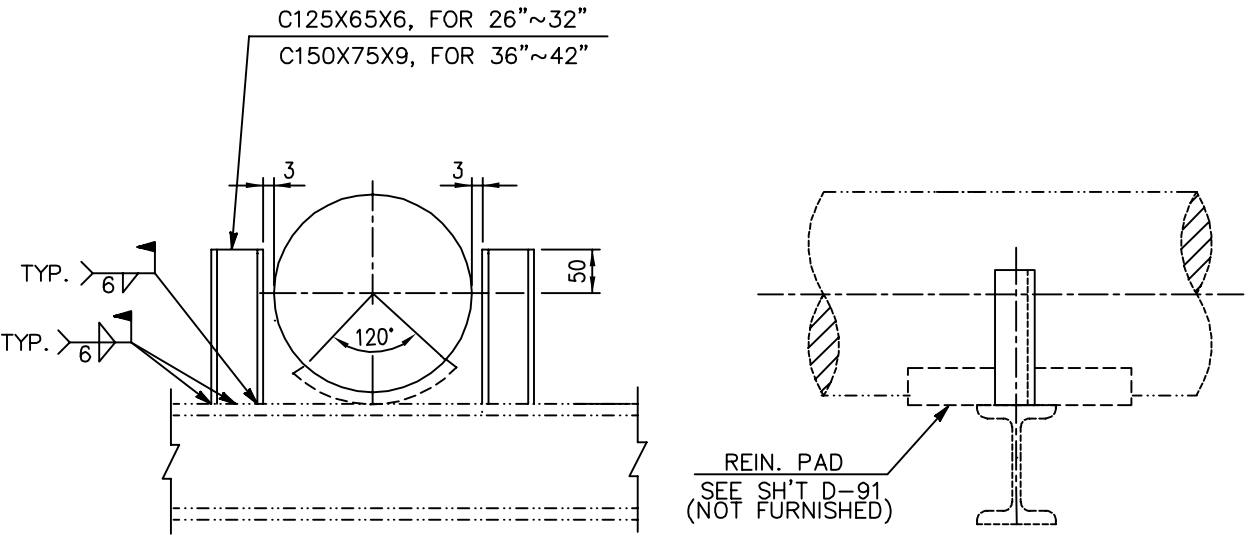
BY		CHK.	
DATE		APPR.	

(1 OF 2)

D-62

REV.
1

FOR PIPE SIZE 26"~42"



ELEV.

NOTES:

1. FOR DESIGNATION NUMBER OF SUPPORT SEE SH'T D-62.



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DETAIL OF PIPE SUPPORT

TYPE-51

(2 OF 2)

ENGINEERING STANDARD

D7TS-701-E

D-62A

REV.
1

BY		CHK.	
DATE		APPR.	

150 (NOTE 2)

DETAIL
SEE D-80

Technical drawing of a roof edge detail. The drawing shows a 120-degree curved roof section. An L40x40x5 angle is cut in the field. The drawing includes dimensions for the angle (3 TYP.), the roof radius (20 TYP.), and the base plate (40). A note indicates a 120-degree angle. A dimension of 6 is also shown.

150 (NOTE 2)

DETAIL
SEE D-80

5



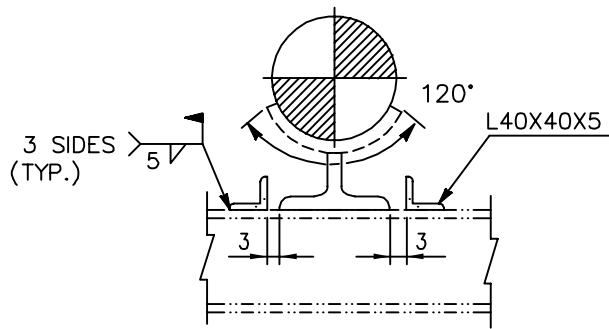
CTCI CORPORATION

REV.
1

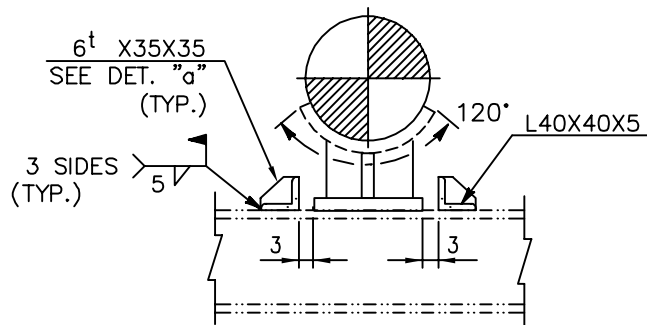
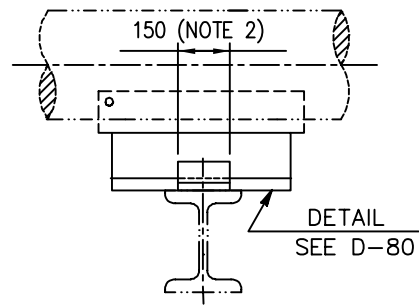
BY
DATE

CHK.
APPR.

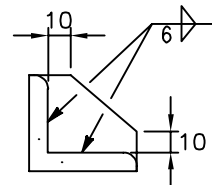
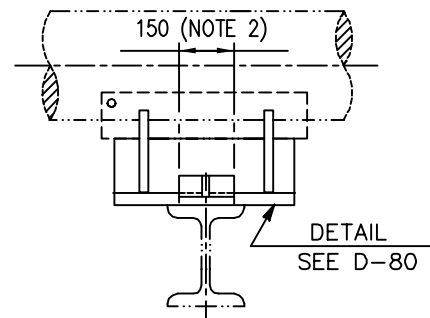
FOR LINE SIZE 3/4" - 24"



(4" & SMALLER)



(6" TO 24" PIPE)



DET. "a"

ELEV.

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

53-2B(P)-A(A)-130-500

DENOTE TYPE NO. ————

DENOTE LINE SIZE ————

DENOTE REIN. PAD IS REQ'D ————

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "B"

SEE SH'T D-80A TABLE "A"

2. THE MAX. LENGTH OF THIS MEMBER SHALL NOT EXCEED THE WIDTH OF BEAM.

3. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-53

(1 OF 3)

ENGINEERING STANDARD

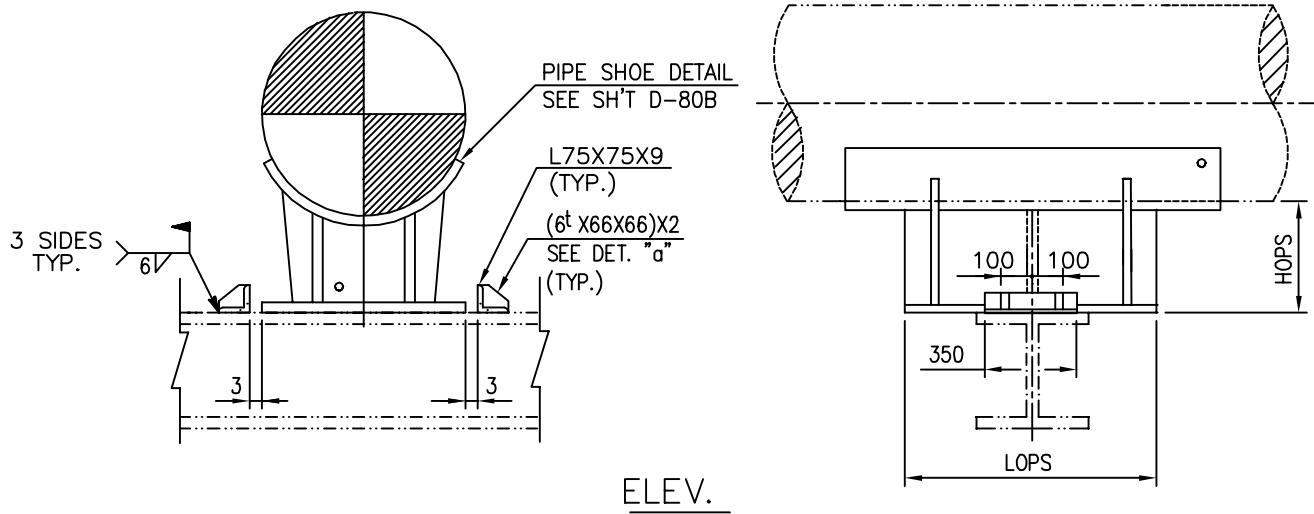
D7TS-701-E

D-64

REV.
1

BY		CHK.	
DATE		APPR.	

FOR LINE SIZE 26"–42"



NOTES :

- FOR DIMENSIONS AND INFORMATION NOT SHOWN ON THIS DWG. SEE SH'T D-80B.
- DESIGNATION NUMBER DENOTE AS FOLLOWS :

53-30B-A(A)-130-500
 DENOTE TYPE NO.
 DENOTE LINE SIZE
 SEE NOTE 3
 MODIFY LOPS=500 (IN mm) IF ANY
 MODIFY HOPS=130 (IN mm) IF ANY
 SEE SH'T D-80A TABLE "B"

- TYPE-A FOR INSULATION THICKNESS ≤ 100
 TYPE-B FOR INSULATION THICKNESS > 100 AND ≤ 150
 TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED
- LOADING CONDITION: CARBON STEEL SADDLE WITH MAX. LINE TEMPERATURE 371°C
- HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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 CTCI CORPORATION

DETAIL OF PIPE SUPPORT

TYPE-53

(2 OF 3)

ENGINEERING STANDARD

D7TS-701-E

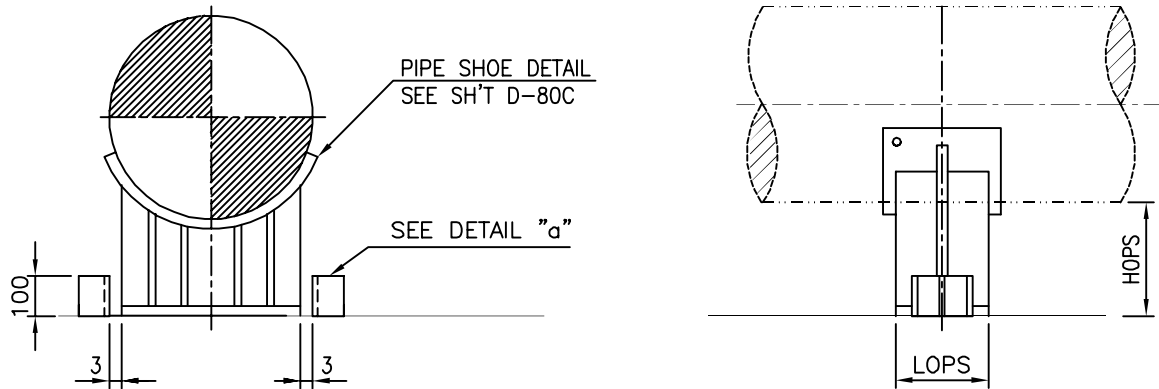
D-64A

 REV.
 1

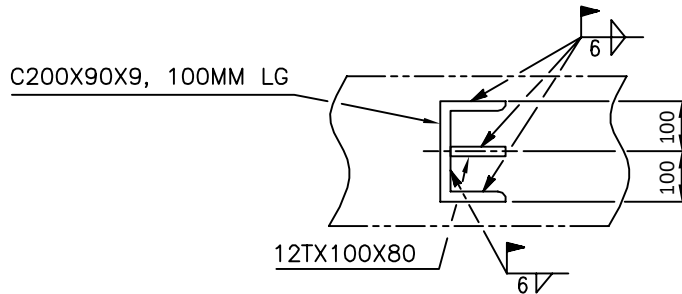
 BY
 DATE

 CHK.
 APPR.

FOR LINE SIZE 44"-78"



ELEV.



DETAIL "a"

NOTES :

1. FOR DIMENSIONS AND INFORMATION NOT SHOWN ON THIS DWG. SEE SH'T D-80C.
2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

53-60B-A(A)-130-600

DENOTE TYPE NO. ☐ MODIFY LOPS=600 (IN mm) IF ANY

DENOTE LINE SIZE ☐ MODIFY HOPS=130 (IN mm) IF ANY

SEE NOTE 3 ☐ SEE SH'T D-80A TABLE "B"

3. TYPE-A FOR INSULATION THICKNESS ≤ 100
TYPE-B FOR INSULATION THICKNESS > 100 AND ≤ 150
TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED.
4. LOADING CONDITION: CARBON STEEL SADDLE WITH MAX. LINE TEMPERATURE 371°C
5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-53

(3 OF 3)

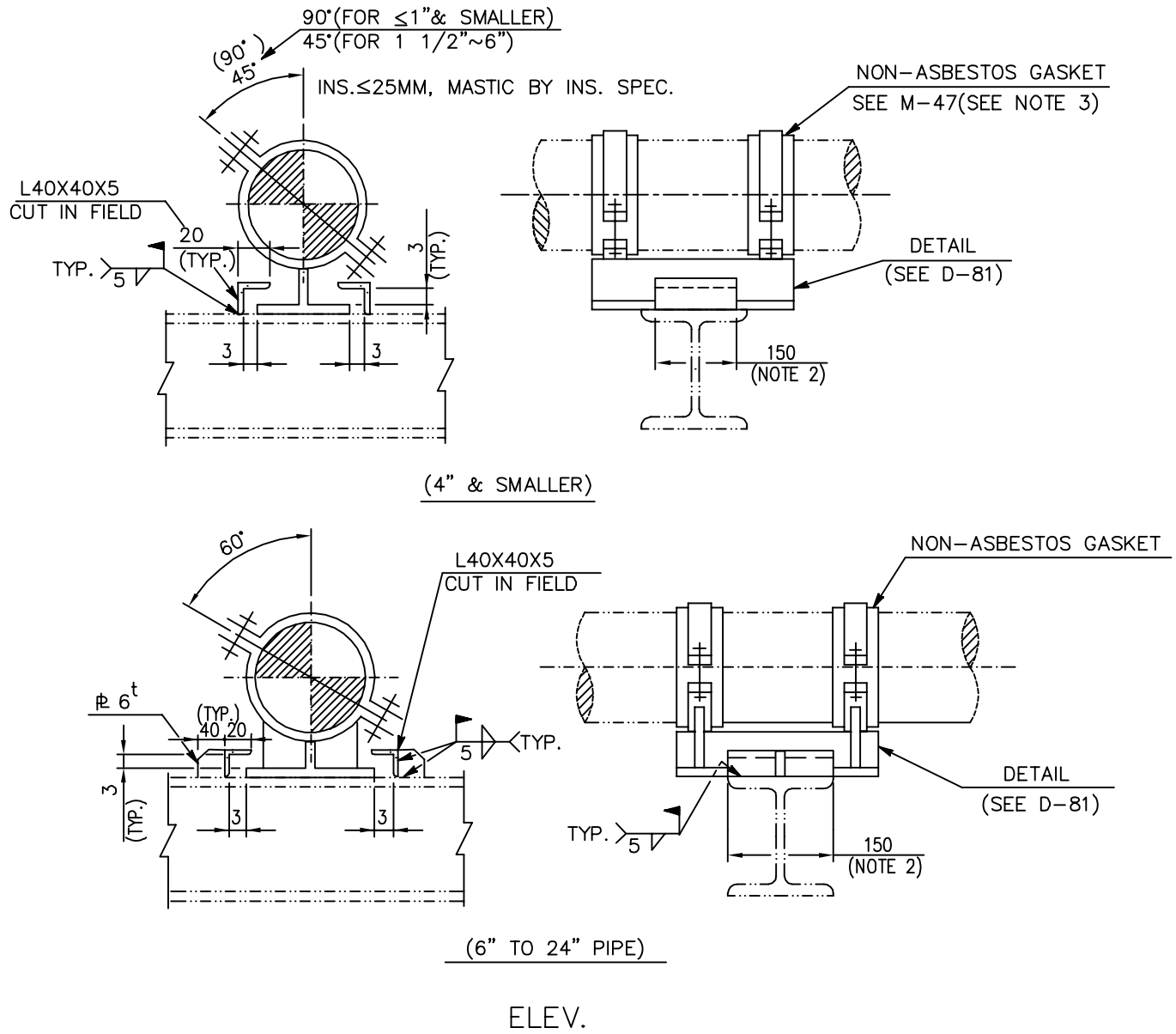
ENGINEERING STANDARD

D7TS-701-E

D-64B

REV.
1

BY		CHK.	
DATE		APPR.	



NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

54-2B-A-130-500

DENOTE TYPE NO.		DENOTE LINE SIZE			MODIFY LOPS=500 (IN mm) IF ANY	MODIFY HOPS=130 (IN mm) IF ANY	SEE SH'T D-80A TABLE "A"
-----------------	--	------------------	--	--	--------------------------------	--------------------------------	--------------------------

2. THE MAX. LENGTH OF THIS MEMBER SHALL NOT EXCEED THE WIDTH OF BEAM.
3. NON-ASBESTOS GASKET IS USED FOR ALLOY STEEL AND STAINLESS STEEL LINES.
(MAX. LINE TEMP.: 371°C)
4. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-54

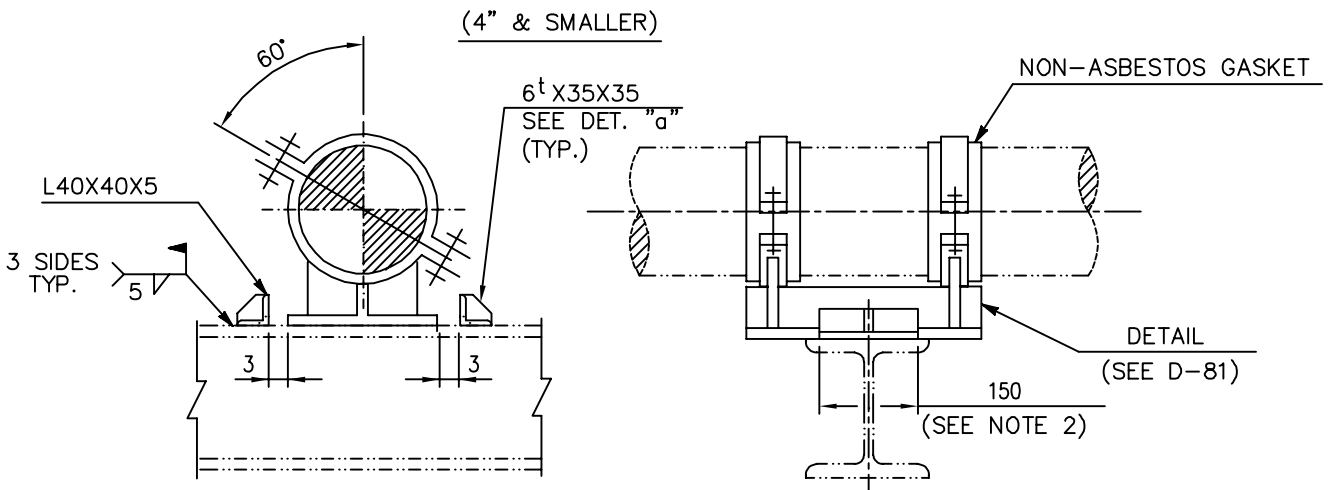
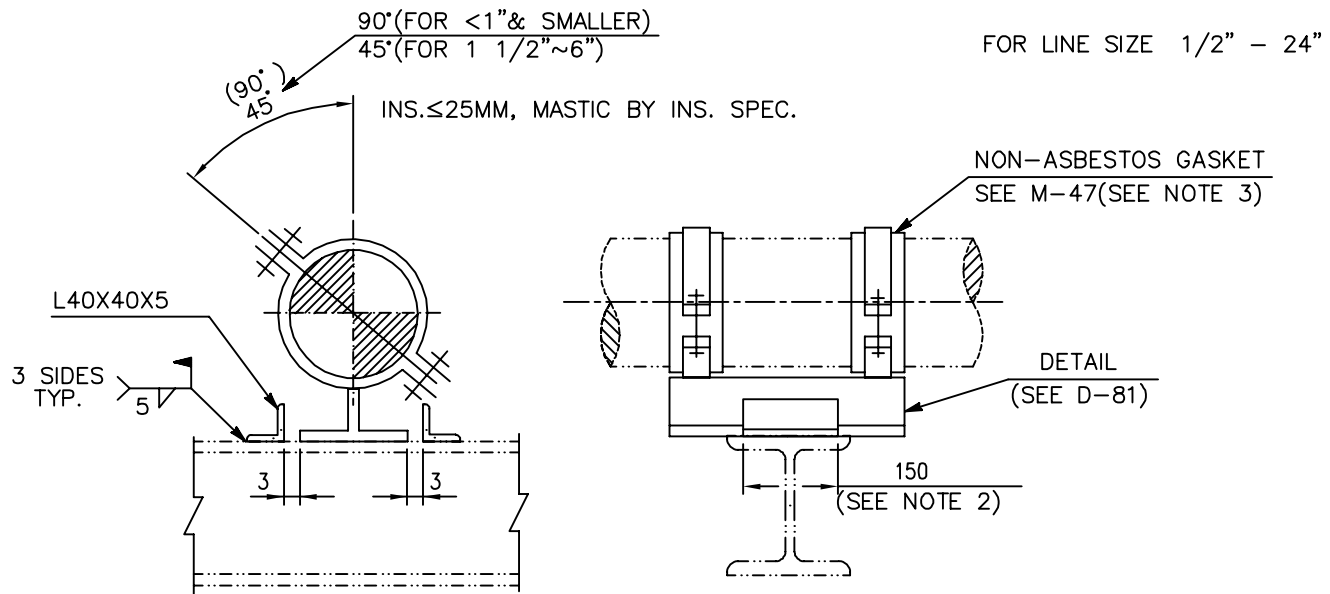
ENGINEERING STANDARD

D7TS-701-E

BY		CHK.	
DATE		APPR.	

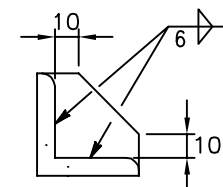
D-65

 REV.
1



(6" TO 24" PIPE)

ELEV.



DET. "a"

NOTES:

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

55-2B-A-130-500

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "A"

DENOTE TYPE NO. DENOTE LINE SIZE

2. THE MAX. LENGTH OF THIS MEMBER SHALL NOT EXCEED THE WIDTH OF BEAM.

3. NON-ASBESTOS GASKET IS USED FOR ALLOY STEEL AND STAINLESS STEEL LINES.
(MAX. LINE TEMP. : 371°C)

4. HOPS=HEIGHT OF SHOE, LOPS=LENGTH OF SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-55

(1 OF 2)

ENGINEERING STANDARD

D7TS-701-E

D-66

REV.
1

BY

DATE

CHK.

APPR.

CLAMPED PIPE SHOE
DETAIL SEE SH'T D-81A

(6" X 66 X 66) X 2
SEE DET. "a"
(TYP.)

3 SIDES
TYP.

6"

L75 X 75 X 9
(TYP.)

3

3

NON-ASBESTOS GASKET
SEE M-47 (SEE NOTE 4)

100

100

350

HOPS

LOPS

ELEV.

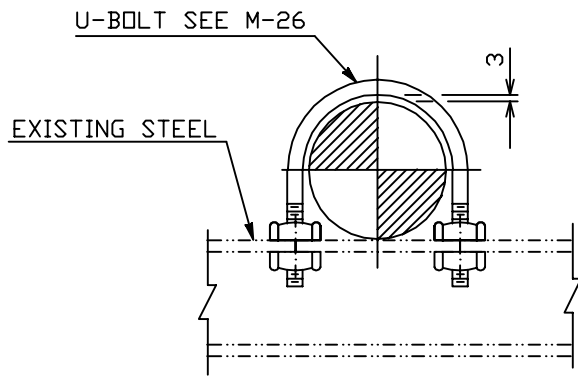


FIG.-A
(SLIDE)

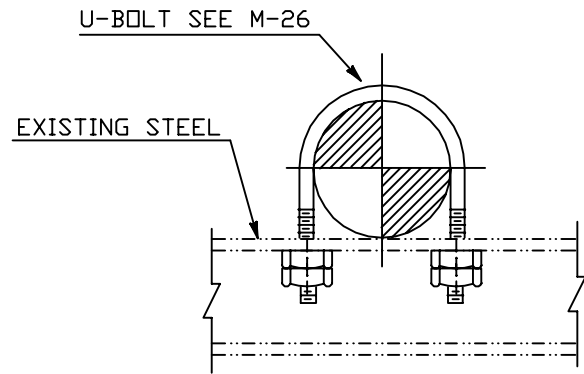


FIG.-B
(FIXED)

ELEV.

LINE SIZE	ROD SIZE	U-BOLT SEE M-26
1/4"	1/4"	UB - 1/4B
1/2"	1/4"	UB - 1/2B
3/4"	1/4"	UB - 3/4B
1"	1/4"	UB - 1B
1 1/4"	3/8"	UB - 1 1/4B
1 1/2"	3/8"	UB - 1 1/2B
2"	3/8"	UB - 2B
2 1/2"	1/2"	UB - 2 1/2B
3"	1/2"	UB - 3B
3 1/2"	1/2"	UB - 3 1/2B
4"	1/2"	UB - 4B
5"	1/2"	UB - 5B
6"	5/8"	UB - 6B
8"	5/8"	UB - 8B
10"	3/4"	UB - 10B
12"	7/8"	UB - 12B
14"	7/8"	UB - 14B
16"	7/8"	UB - 16B
18"	1"	UB - 18B
20"	1"	UB - 20B
24"	1"	UB - 24B
30"	1"	UB - 30B

NOTES:

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

57 - 2B -A(S)

DENOTE FOR S.S. U-BOLT IF REQUIRED
 DENOTE FIG. NO.
 DENOTE LINE SIZE
 DENOTE TYPE NO.



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DETAIL OF PIPE SUPPORT
TYPE-57

ENGINEERING STANDARD

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D-68

REV.
1

BY		CHK.	
DATE		APPR.	

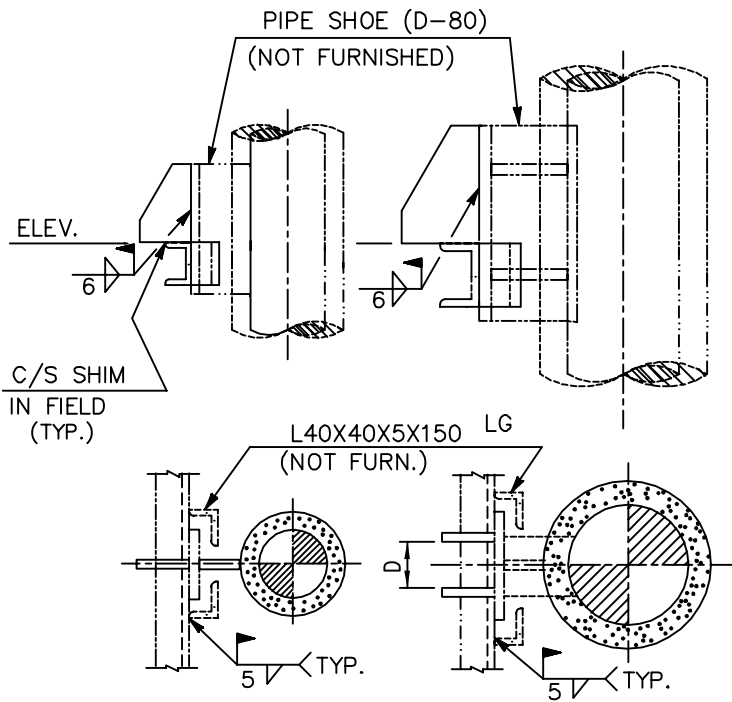


FIG. -A
(FOR INS. PIPE)

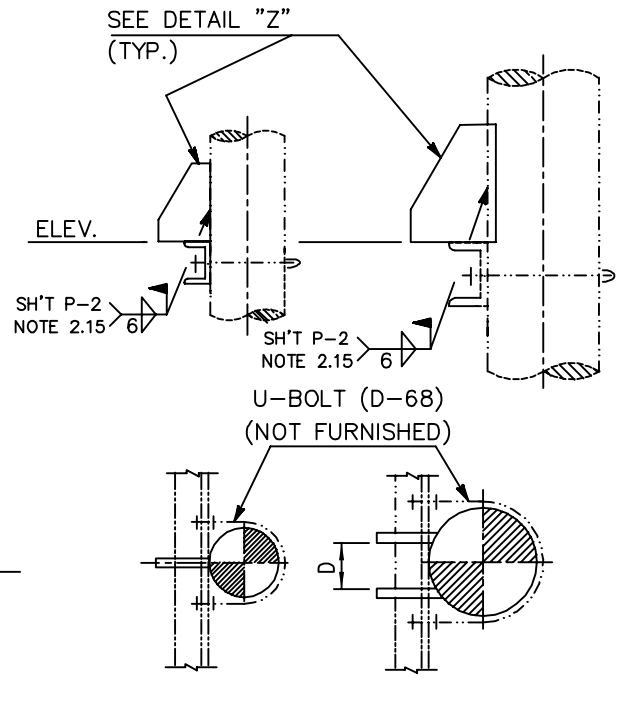
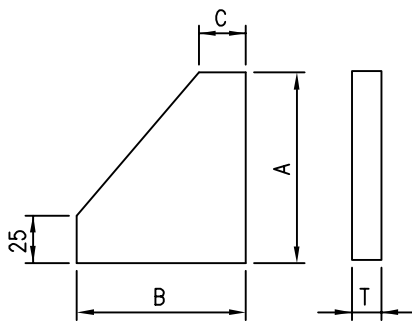


FIG. -B
(FOR BARE PIPE)



DETAIL "Z"

MAT'L OF MAIN PIPE	MAT'L OF SUPPORT	SYMBOL
CARBON STEEL	A-283-C	NONE
ALLOY STEEL	A387-22	(A)
STAINLESS STEEL	A240-304	(S)
CARBON STEEL (A516-60)	A516-60	(R)
ALLOY-825	B424-N08825-S	(N)

TABLE "A"

DIMENSIONS (mm)

LINE SIZE	A	B	C	D	T	T (FOR STAINLESS STEEL ONLY)
2 1/2" & SMALLER	80	55	15	-	9	6
3"-8"	150	100	50	-	12	9
10"-14"	150	130	50	120	12	-

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

59-14B-B(S)
 — SEE TABLE "A" (ONLY USED WITH FIG. B)
 — DENOTE FIG. NO.
 — DENOTE LINE SIZE
 — DENOTE TYPE NO.

2. MATERIAL : SEE TABLE "A"



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DETAIL OF PIPE SUPPORT

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ENGINEERING STANDARD

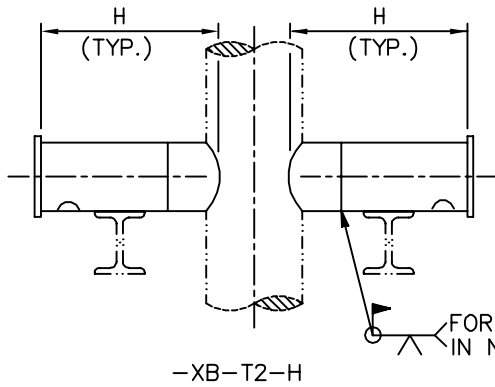
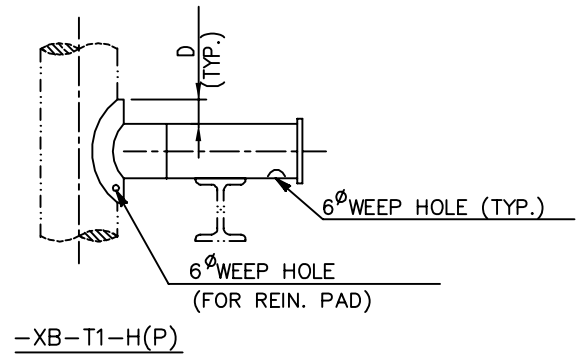
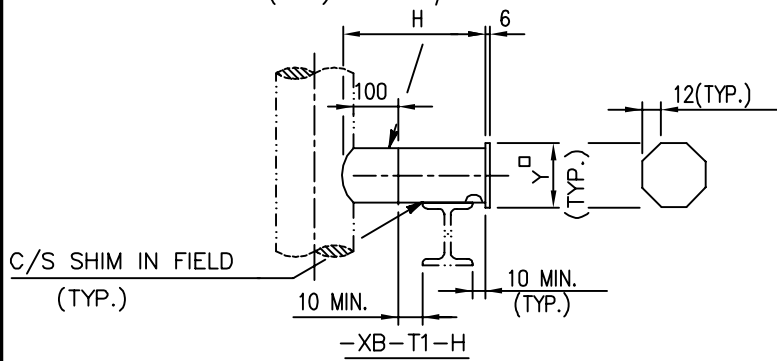
D7TS-701-E

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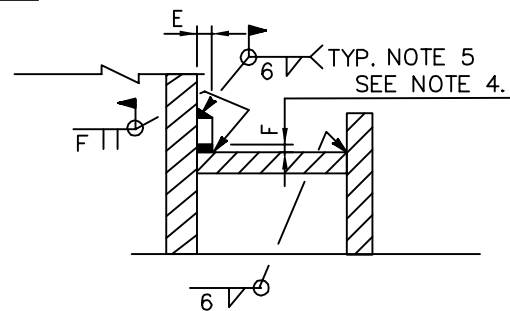
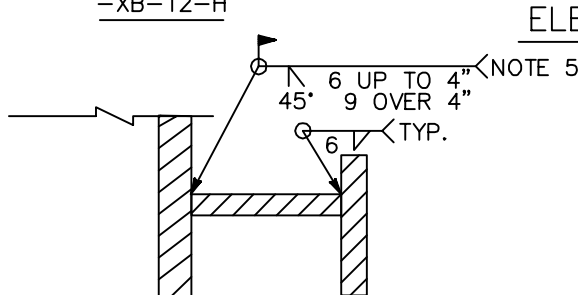
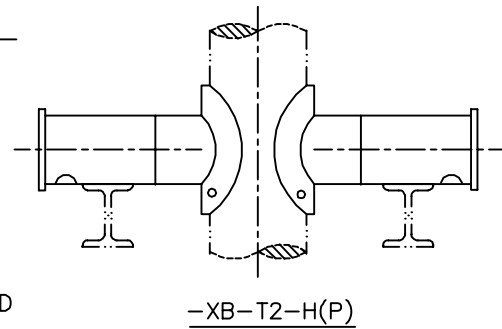
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DATE		APPR.	

FOR SPECIAL MAIN LINE
SEE NOTE 2 (TYP.)



ELEV.



WELD DETAIL
(NO PAD REQ'D)

WELD DETAIL
(PAD REQ'D)

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

61-3B-T1-05(P4)

DENOTE TYPE NO. ———— DENOTE MAIN LINE SIZE
DENOTE NOM. TRUNN. PIPE SIZE ———— DENOTE REIN. PAD IS REQ'D.
————— DENOTE THE LG. OF TRUNN. (IN 100mm)
————— DENOTE ONE-TRUNN. TYPE

2. FOR ALLOY STEEL, STAINLESS STEEL & STRESS RELIEF LINES, THE MATERIALS SHALL BE THE SAME AS MAIN LINES, AND IT SHALL BE FABRICATED TOGETHER WITH MAIN LINES IN SHOP.
3. THE MOMENT CAPACITY OF THE TABLE ARE FOR ONE-TRUNN. TYPE, FOR TWO-TRUNN. TYPE THE CAPACITIES SHALL BE DOUBLED.
4. DIMENSION "F" : ALLOY OR STAINLESS TAKES 6 mm
CARBON STEEL TAKES 9 mm
5. ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT

TYPE-61

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FOR CARBON STEEL

CORRECTION FACTORS								
TEMP. °F	200	300	400	500	600	650	700	750
FACTOR	1.000	1.055	1.109	1.168	1.239	1.275	1.313	1.430

DIMENSION (mm)

		TRUNNION – NOMINAL PIPE SIZE																	
		2"		3"		4"		6"		8"		10"		12"		14"			
		SCH. 80		SCH. 40 OR GREATER										3/8"W OR GREATER					
Y		80		110		140		190		240		290		340		380			
NOM. PIPE SIZE	PIPE SCH.	PAD REQUIREMENTS																PAD TH'K MIN. "E"	DIM. "D" mm
		NO PAD		NO PAD		NO PAD		NO PAD		NO PAD		NO PAD		NO PAD		NO PAD			
		MOMENT CAPACITY OF ONE TRUNNION. (INCH–KIPS)																	
3"	40	1	6															5.5	40
4"	40	1	7	3	15													6.0	50
6"	40	2	8	4	17	6	28											7.0	65
8"	40	2	9	5	19	8	32	17	69									8.5	90
10"	40			6	21	9	35	20	77	34	130							9.0	100
12"	3/8"W			5	21	9	34	19	74	33	125	51	195					9.5	100
14"	3/8"W					9	33	19	71	31	120	49	186	69	262			9.5	115
16"	3/8"W					8	30	17	66	29	112	46	174	64	245	78	295	9.5	115
18"	3/8"W							16	62	28	106	43	164	61	231	73	278	9.5	125
20"	3/8"W							16	59	26	100	41	156	57	219	69	264	9.5	125
24"	3/8"W							14	54	24	91	37	142	52	200	63	241	9.5	140
30"	3/8"W									21	82	33	127	47	179	57	215	9.5	165
36"	3/8"W											30	116	43	163	52	197	9.5	180
42"	3/8"W											28	107	40	151	48	182	9.5	190
3"	80	3	9															5.5	40
4"	80	3	10	7	21													6.0	50
6"	80	5	11	10	26	17	43											7.0	65
8"	80	5	11	12	26	20	49	43	106									8.5	90
10"	1/2"W			11	26	18	48	38	104	65	175							9.0	100
12"	1/2"W			10	26	16	45	35	97	59	164	92	255					9.5	100
14"	1/2"W					15	43	33	93	57	157	88	244	124	343			9.5	115
16"	1/2"W					14	40	31	87	53	147	82	228	116	321	140	387	9.5	115
18"	1/2"W							30	82	50	138	78	215	109	302	132	365	9.5	125
20"	1/2"W							28	77	47	131	74	204	104	287	125	346	9.5	125
24"	1/2"W							26	71	43	120	67	186	95	262	114	316	9.5	140
30"	1/2"W									39	107	60	167	85	234	102	283	9.5	165
36"	1/2"W											55	152	77	214	93	258	9.5	180
42"	1/2"W											51	141	72	198	86	239	9.5	190



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DETAIL OF PIPE SUPPORT

TYPE-61

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FOR STAINLESS STEEL

TEMP. °F

CORRECTION FACTORS											
TEMP.	TP304	TP316	TP347	TEMP.	TP304	TP316	TP347	TEMP.	TP304	TP316	TP347
-325 TO				750	1.60	0.99	1.13	1150	2.90	1.96	2.08
200	1.00	0.89	0.89	800	1.67	0.99	1.14	1200	3.70	2.45	3.33
300	1.11	0.93	0.98	850	1.72	1.01	1.16	1250	5.12	3.14	4.63
400	1.22	0.95	1.05	900	1.77	1.04	1.18	1300	6.80	4.16	6.17
500	1.33	0.97	1.10	950	1.83	1.10	1.20	1350	9.25	5.55	8.32
600	1.44	0.97	1.12	1000	1.89	1.19	1.23	1400	11.89	7.09	10.74
650	1.49	0.98	1.12	1050	1.96	1.36	1.27	1450	16.65	9.00	13.88
700	1.54	0.98	1.13	1100	2.22	1.60	1.33	1500	22.20	11.10	16.65

DIMENSION (mm)

		TRUNNION – NOMINAL PIPE SIZE																	
		2”		3”		4”		6”		8”		10”		12”		14”			
		MIN. SCH. SAME AS HEADER																	
Y		80		110		140		190		240		290		340		380			
NOM. PIPE SIZE	PIPE SCH.	PAD REQUIREMENTS																PAD TH’K MIN. ”E”	DIM. ”D”
		NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD	NO PAD		
		MOMENT CAPACITY OF ONE TRUNNION. (INCH–KIPS)																	
3”	5S	1	3															2.0	30
4”	5S	1	2	2	5													2.0	40
6”	5S	1	3	2	6	4	10											3.0	45
8”	5S	1	3	2	5	3	9	7	20									3.0	60
10”	5S			2	7	4	11	8	24	14	41							3.5	65
12”	5S			2	8	5	14	11	30	18	51	28	79					4.0	75
14”	5S					4	12	9	26	16	45	25	70	35	98			4.0	75
16”	5S					4	12	10	27	16	46	25	71	35	100	42	120	4.0	85
18”	5S							9	25	15	43	24	67	33	94	40	113	4.0	90
20”	5S							10	29	18	50	27	77	39	108	47	131	5.0	100
24”	5S							12	33	20	57	31	88	44	124	53	149	5.5	120
30”	5S									22	62	34	97	48	136	58	164	6.5	145
3”	10S	2	5															3.0	30
4”	10S	1	4	3	9													3.0	40
6”	10S	1	4	3	9	5	14											3.5	45
8”	10S	1	4	3	9	5	14	11	31									4.0	60
10”	10S			3	9	5	15	12	33	20	55							4.0	65
12”	10S			3	10	6	16	12	34	21	58	32	90					4.5	75
14”	10S					6	16	12	35	21	59	33	92	46	130			5.0	75
16”	10S					5	15	12	33	20	55	30	86	43	121	52	146	5.0	85
18”	10S							11	31	18	52	29	81	40	114	49	138	5.0	90
20”	10S							13	37	22	62	34	96	48	135	58	163	5.5	100
24”	10S							14	41	25	70	38	108	54	152	65	183	6.5	120
30”	10S									31	87	48	135	67	189	81	228	8.0	145



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DETAIL OF PIPE SUPPORT

TYPE-61

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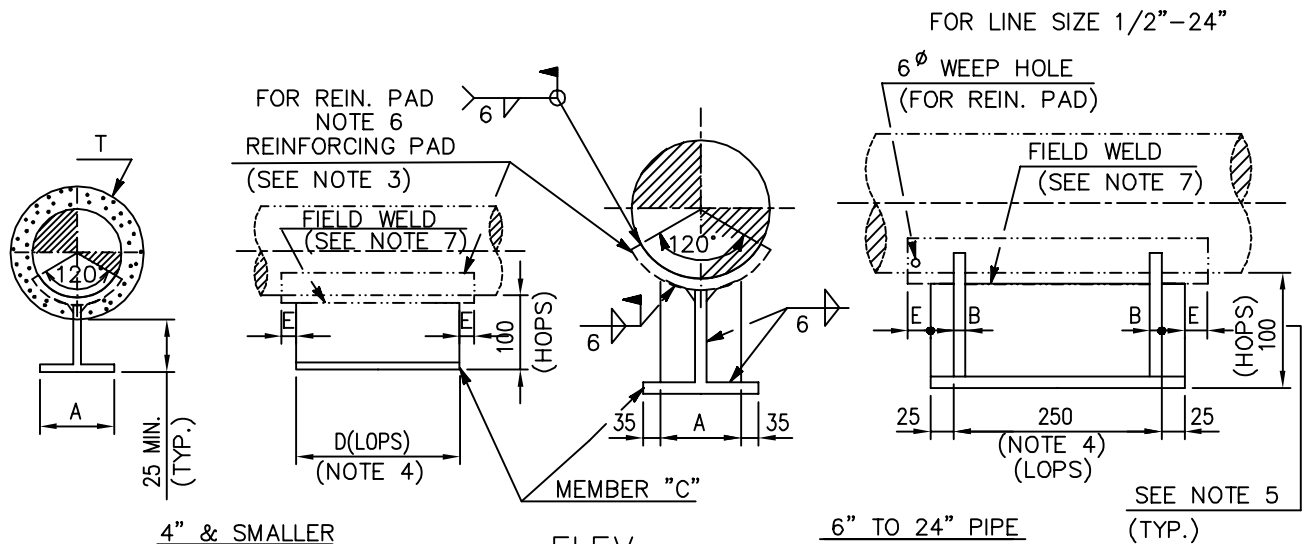
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DIMENSION (mm)

LINE SIZE	A	B	C (NOTE 1)	D (NOTE 4)	E
1/2"	100	—	CUT FROM H-200X100X5.5X8	150	—
3/4"	100	—		150	—
1"	100	—		150	—
1 1/2"	100	—		150	—
2"	100	—		150	50
2 1/2"	100	—		150	50
3"	100	—		250	50
4"	100	—		250	50
5"	100	—		250	50
6"	130	9	CUT FROM H-200X200X8X12	—	50
8"	130	9		—	50
10"	130	9		—	50
12"	130	9		—	50
14"	130	9		—	50
16"	250	12	FAB. FROM 12 ^t PLATE	—	50
18"	250	12		—	50
20"	250	12		—	50
24"	300	12		—	50

NOTES : (FOR SH'T D-80,80A ONLY)

1. SHOES FOR PIPES 1 1/2" & SMALLER MAY BE FABRICATED FROM 6^t PLATE, 2" THRU. 14" FROM 9^t PLATE, 16" THRU. 24" FROM 12^t PLATE

2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

66-1 1/2B(P)-A(A)-130-500

DENOTE TYPE NO. —

DENOTE LINE SIZE —

DENOTE REIN. PAD IS REQ'D —

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "A" —

SEE SH'T D-80A TABLE "B"

3. FOR PADS MATERIAL USE PIPE MATERIAL THAT CUT FROM THE PIPE OR EQUIVALENT TO THE PIPE.

4. LONGITUDINAL LENGTH OF PIPE SHOE (SAY LOPS)

SHALL BE CALCULATED IN THE CASE WITH PIPE STOP OR THERMAL MOVEMENT.



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DETAIL OF PIPE SUPPORT

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NOTES : (CONT'D)

5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.

TABLE "A"

(FOR NORMAL INSULATION LINE)

INSULATION TH'K (mm)	HEIGHT OF SHOE (mm) (SAY HOPS)	SYMBOL
75 & LESSER	100	NONE
80 THRU. 125	150	A
130 THRU. 175	200	B
180 THRU. 225	250	C

(FOR TRACING INSULATION LINE)

INSULATION TH'K (mm)	HEIGHT OF SHOE (mm) (SAY HOPS)	SYMBOL
50 & LESSER	100	NONE
55 THRU. 100	150	A
105 THRU. 150	200	B
155 THRU. 200	250	C

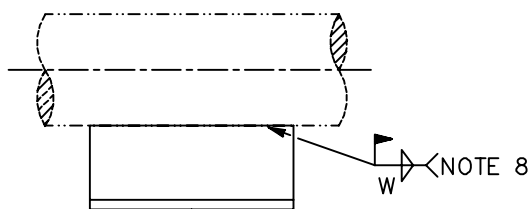
TABLE "B"

MAT'L OF SHOES	FABRICATED FROM (NOTE 1)	SYMBOL
CARBON STEEL	SHAPE STEEL OR CS PLATE	NONE
ALLOY STEEL	AS PLATE	(A)
STAINLESS STEEL	SS PLATE	(S)
LOWER-TEMPERATURE CARBON STEEL	A516-60 PLATE	(R)
ALLOY-825	B424-N08825-S	(N)

6. FOR MATERIAL SPECIFICATION SEE SH'T P-2.

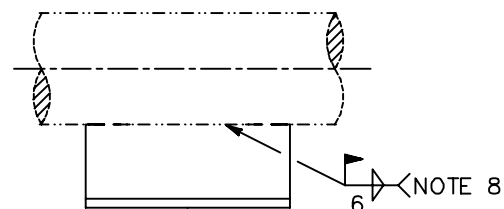
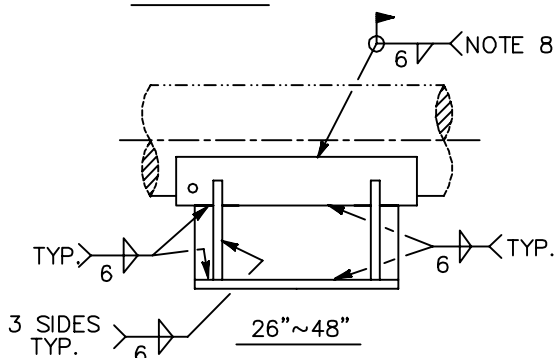
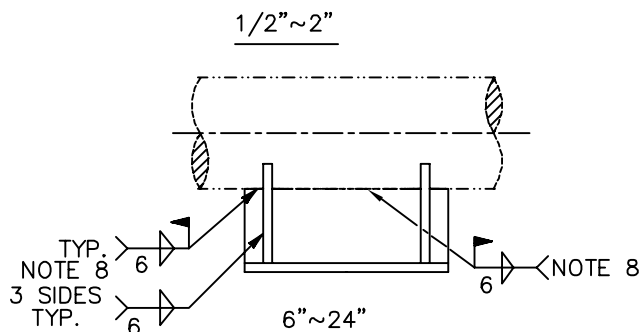
7. FIELD WELD OF PIPE SHOE .

8. ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



W=4 FOR PIPE SCH. 80 & BELOW

W=6 FOR PIPE SCH. 160 & ABOVE

 $2\frac{1}{2}'' \sim 4''$ 

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DETAIL OF PIPE SUPPORT

TYPE-66

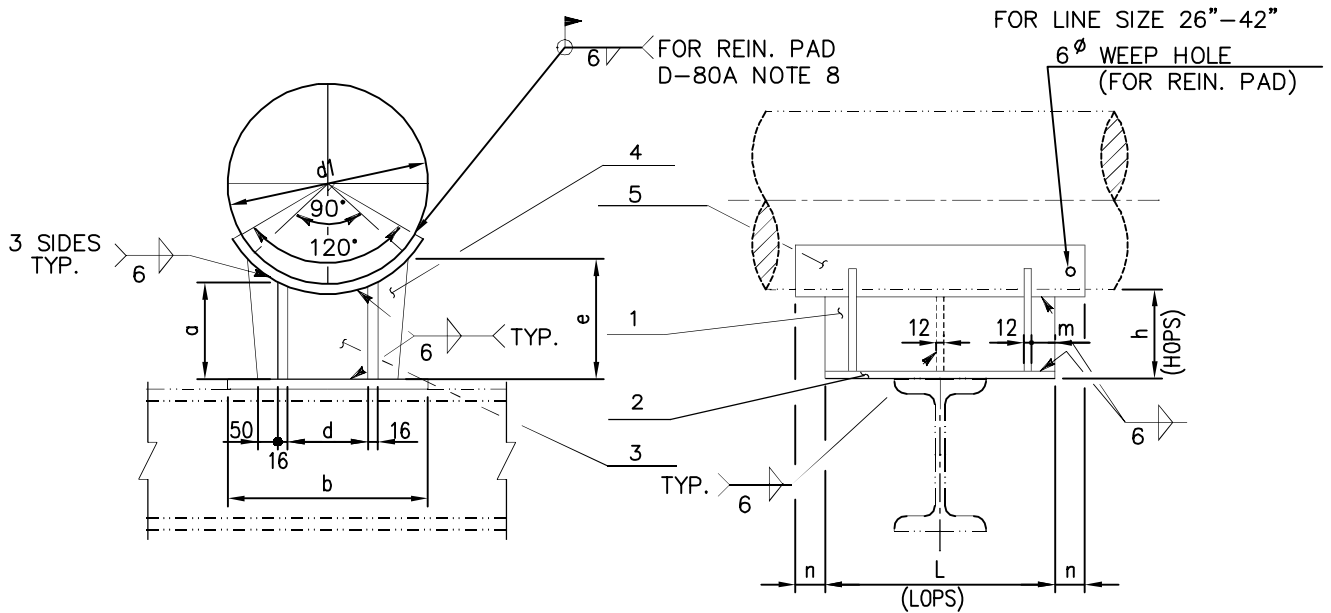
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(FOR OTHER WELDS, SEE SH'T D-80A NOTE 7)

DIMENSION (mm)

NO	1			2	3	4	5											
PIECE	2			1	1	4	1											
PIPE SIZE	(NOTE 4) FOOT a			FOOT PLATE	PLATE	PLATE	PAD	DIMENSION										
	A	B	C					b	d	e (NOTE 2,4)			h (NOTE 2)			L	m	n
	(NOTE 2)	(NOTE 2)	(NOTE 2)							A	B	C	A	B	C			
26"	148	198	298	16 TH'K	12 TH'K	12 TH'K	NOTE 6	430	230	222	272	372	150	200	300	540	35	50
28"	148	198	298					440	240	230	280	380	150	200	300	540	35	50
30"	148	198	298					450	250	237	287	387	150	200	300	540	35	50
32"	148	198	298					460	260	244	294	394	150	200	300	540	35	50
34"	150	200	300					480	280	252	302	402	150	200	300	540	35	50
36"	152	202	302					500	300	259	309	409	150	200	300	540	35	50
40"	156	206	306					535	335	274	324	424	150	200	300	540	35	50
42"	156	206	306					550	350	281	331	431	150	200	300	540	35	50

NOTES : (FOR SH'T D-80B ONLY)

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

66-28B-A(A)-130-500

DENOTE TYPE NO.

DENOTE LINE SIZE

SEE NOTE 2

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "B"

2. TYPE-A FOR INSULATION THICKNESS ≤ 100 TYPE-B FOR INSULATION THICKNESS > 100 AND ≤ 150 TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED

3. LOADING CONDITION: CARBON STEEL SADDLE WITH MAX. LINE TEMPERATURE 371°C

4. DIMENSION "a" & "e" ARE CALCULATED BASE ON PIPE SHOE WITH 12 mm TH'K REINF. PAD.

5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE

6. REINFORCING PAD MATERIALS ARE CUT FROM PIPES.



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DETAIL OF PIPE SUPPORT

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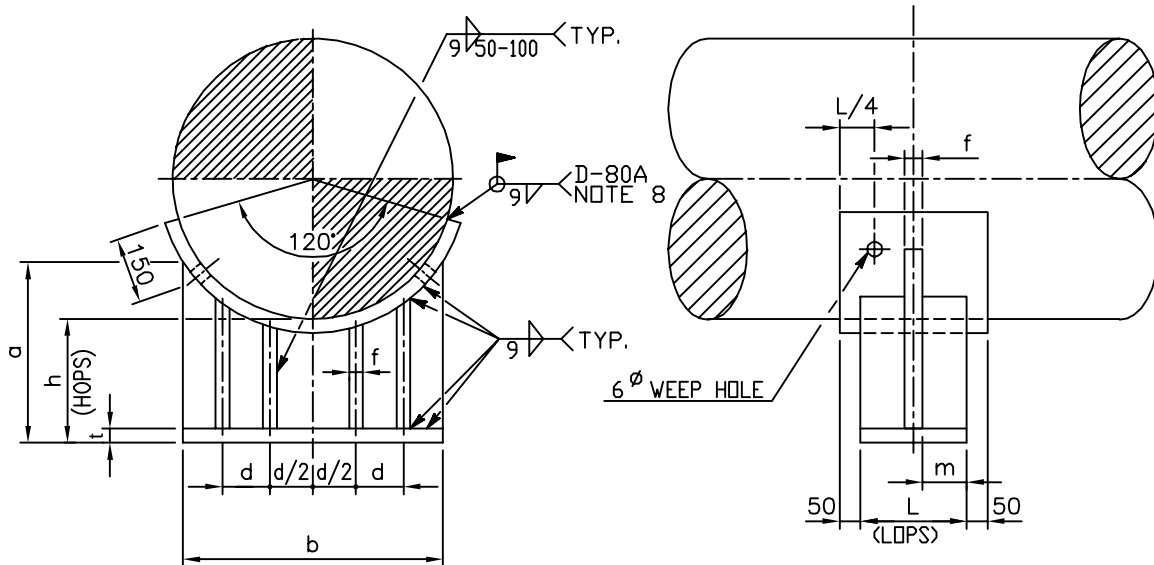
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FOR LINE SIZE 44"-78"



PIPE SIZE	a			b	t	d	h			PIPE SHOE			R. PAD
	A	B	C				A	B	C	f	L	m	
44"	376	426	576	950	16	260	150	200	300	16	600	292	NOTE 6
46"	358	408	508	950	16	260							
48"	374	424	524	1000	16	280							
54"	390	440	540	1100	16	310							
56"	407	457	557	1150	16	320							
60"	404	454	554	1200	19	330							
64"	436	486	586	1300	19	360							
66"	462	512	612	1300	19	360							
72"	469	519	619	1450	19	410							
76"	501	551	651	1550	19	440							
78"	536	586	686	1600	19	450							

NOTES : < FOR SH'T D-80C ONLY>

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

66-60B-A(A)-130-600

DENOTE TYPE NO. _____

DENOTE LINE SIZE _____

SEE NOTE 2 _____

MODIFY LOPS=600 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "B"

2. TYPE-A FOR INSULATION THICKNESS ≤ 100 TYPE-B FOR INSULATION THICKNESS $> 100 \leq 150$ TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED

3. LOADING CONDITION: CARBON STEEL SADDLE WITH MAX. LINE TEMPERATURE 371°C

4. DIMENSION "a" IS CALCULATED BASE ON PIPE SHOE WITH 12 mm TH'K REINF. PAD.

5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE

6. REINFORCING PAD MATERIALS ARE CUT FROM PIPES.



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DETAIL OF PIPE SUPPORT

TYPE-66

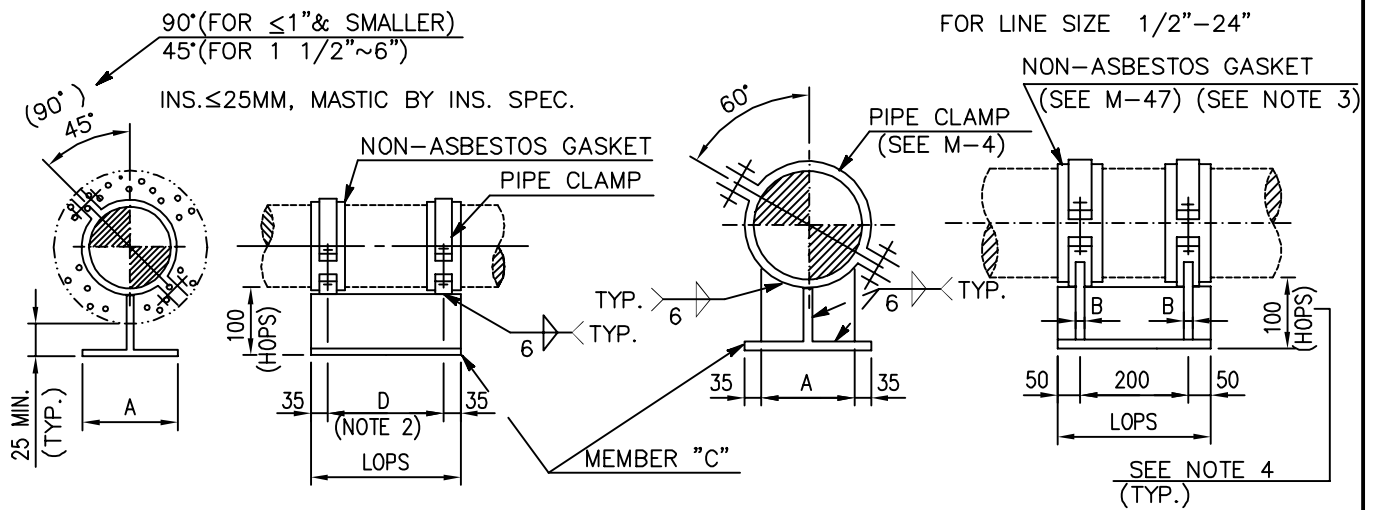
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ENGINEERING STANDARD

D7TS-701-E

D-80C

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(4" & SMALLER)

ELEV.

(6" TO 24" PIPE)

DIMENSIONS (mm)

LINE SIZE	A	B	C	PIPE CLAMP TYPE	D ⁽²⁾
1/2"	100	—	CUT FROM H200X100X5.5X8	PCL-A-1/2B	80
3/4"	100	—		PCL-A-3/4B	80
1"	100	—		PCL-A- 1B	80
1 1/2"	100	—		PCL-A-1 1/2B	80
2"	100	—		PCL-A- 2B	80
2 1/2"	100	—		PCL-A-2 1/2B	80
3"	100	—		PCL-A- 3B	180
4"	100	—		PCL-A- 4B	180
5"	100	—		PCL-A- 5B	180
6"	130	9	CUT FROM H200X200X8X12	PCL-A- 6B	—
8"	130	9		PCL-A- 8B	—
10"	130	9		PCL-A- 10B	—
12"	130	9		PCL-A- 12B	—
14"	130	9		PCL-A- 14B	—
16"	250	12	FAB. FROM 12 ^t PLATE	PCL-A- 16B	—
18"	250	12		PCL-A- 18B	—
20"	250	12		PCL-A- 20B	—
24"	300	12		PCL-A- 24B	—

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

67-3/4B-A-130-500

DENOTE TYPE NO.

DENOTE LINE SIZE

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE NOTE 4

2. LONGITUDINAL LENGTH OF PIPE SHOE (SAY LOPS) SHALL BE CALCULATED IN CASE WITH PIPE STOP.

3. NON-ASBESTOS GASKET IS USED FOR ALLOY STEEL AND STAINLESS STEEL LINES.
(MAX. LINE TEMP. : 371°C)

4. REF. SH'T D-80A TABLE "A"

5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



中鼎工程股份有限公司

CTCI CORPORATION

DETAIL OF PIPE SUPPORT

TYPE-67

(1 OF 2)

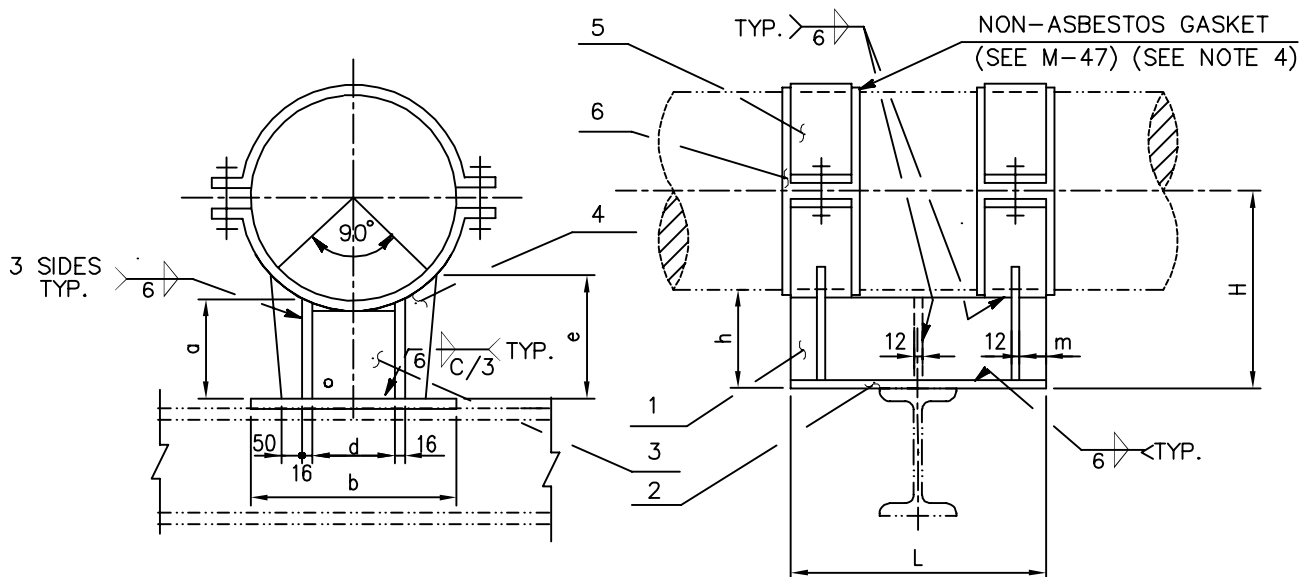
ENGINEERING STANDARD

D7TS-701-E

D-81

REV.
1BY
DATECHK.
APPR.

FOR LINE SIZE 26"—42"



DIMENSIONS (mm)

NO.	1			2	3	4	5	6	DIMENSION												
PIECE	2			1	1	4	2	2													
PIPE SIZE	FOOT a			FOOT PLATE	PLATE	PLATE	CLAMP (NOTE3)	NON-ASBESTOS GASKET													
	A (NOTE 2)	B (NOTE 2)	C (NOTE 2)						b	d	e (NOTE 2)			h (NOTE 2)			H (NOTE 2)			L	m
											A	B	C	A	B	C	A	B	C		
26"	148	198	298	16 TH'K	12 TH'K	12 TH'K	FAB. FROM 12t LP	1.5t TH'K	430	230	222	272	372	150	200	300	480	530	630	540	35
28"	148	198	298						440	240	230	280	380	150	200	300	506	556	656	540	35
30"	148	198	298						450	250	237	287	387	150	200	300	531	581	681	540	35
32"	148	198	298						460	260	244	294	394	150	200	300	556	606	706	540	35
36"	152	202	302						500	300	259	309	409	150	200	300	607	657	757	540	35
40"	156	206	306						535	335	274	324	424	150	200	300	658	708	808	540	35
42"	156	206	306						550	350	281	331	431	150	200	300	683	733	833	540	35

NOTES : (FOR SH'T D-81A ONLY)

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

67-30B-A-130-500

DENOTE TYPE NO. ————
 DENOTE LINE SIZE ————
 ———— MODIFY LOPS=500 (IN mm) IF ANY
 ———— MODIFY HOPS=130 (IN mm) IF ANY
 ———— SEE NOTE 2

2. TYPE-A FOR INSULATION THICKNESS ≤ 100 TYPE-B FOR INSULATION THICKNESS > 100 AND ≤ 150 TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED

3. FOR CLAMP DETAIL SEE SH'T M-56.

4. NON-ASBESTOS GASKET IS USED FOR ALLOY STEEL AND STAINLESS STEEL LINES.
(MAX. LINE TEMP. : 371°C)

5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-67

(2 OF 2)

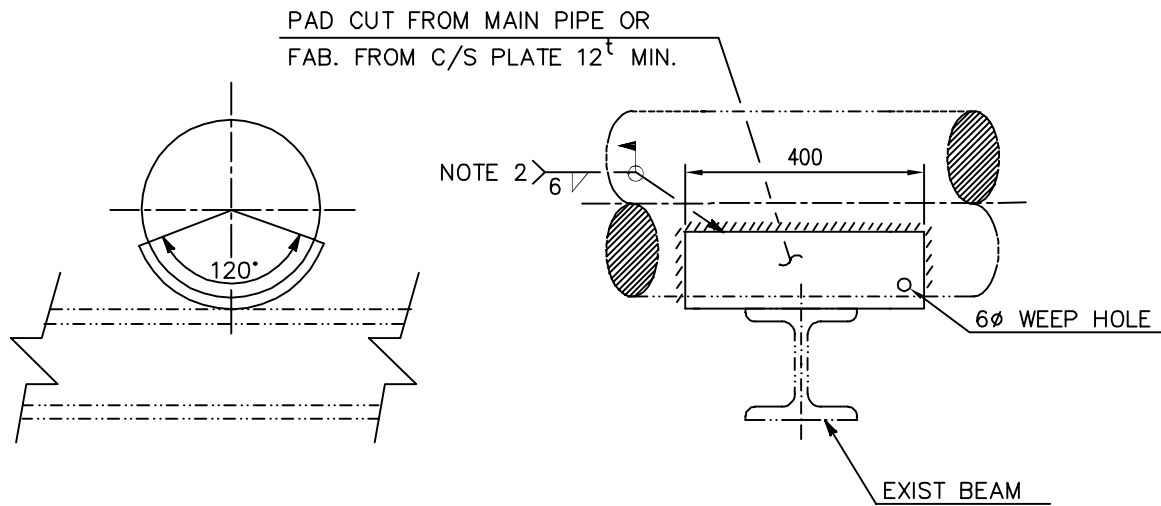
ENGINEERING STANDARD

D7TS-701-E

D-81A

REV.
1BY
DATECHK.
APPR.

FOR LINE SIZE 26" ~ 42"



NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:

76 - 30B

└── DENOTE LINE SIZE

└── DENOTE TYPE NO.

2. ALL WELDED CONNECTIONS SHALL BE MADE WITH 6mm FILLET WELDS, BUT FOR ALL WELDS WITH PIPE, THE WELDING THICKNESS SHALL NOT EXCEED THE THICKNESS OF PIPE UNLESS OTHERWISE NOTED.



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DETAIL OF PIPE SUPPORT
TYPE-76

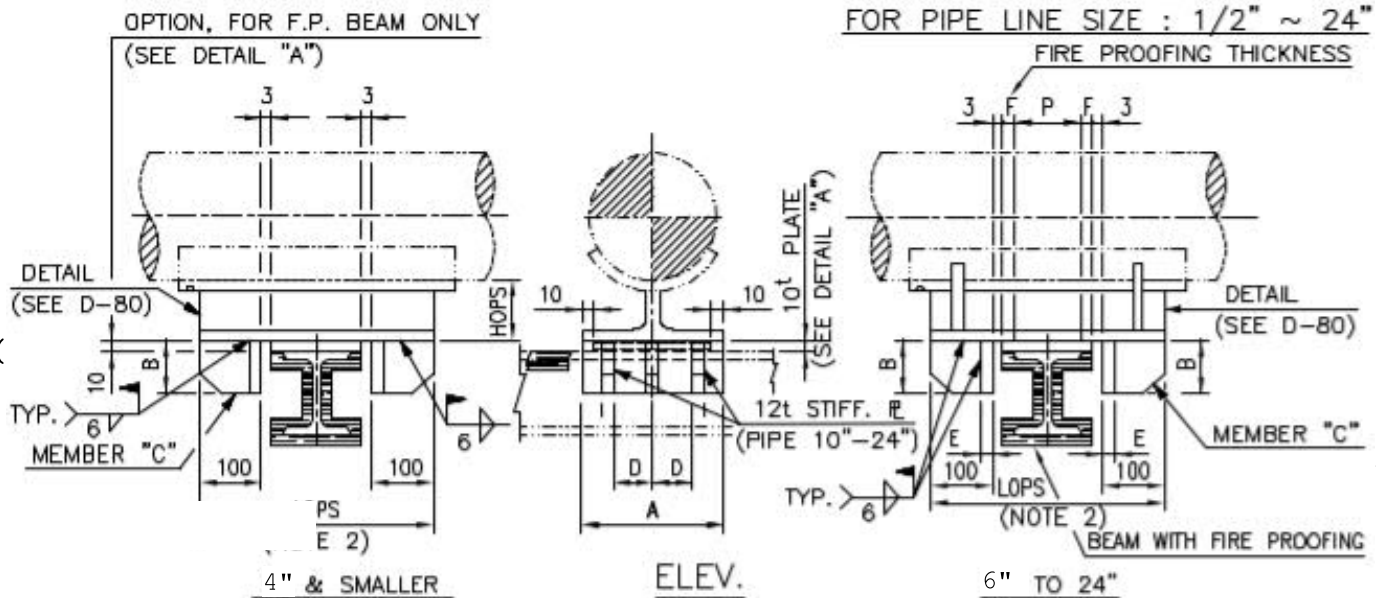
ENGINEERING STANDARD

D7TS-701-E

D-91

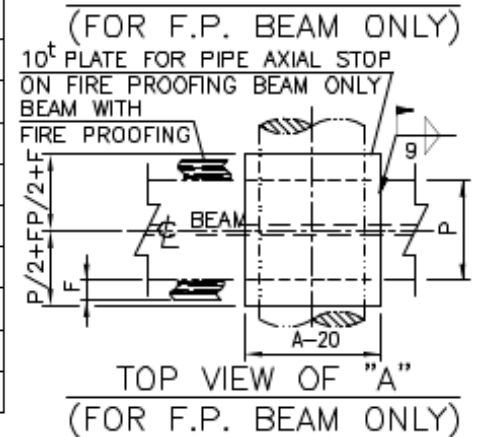
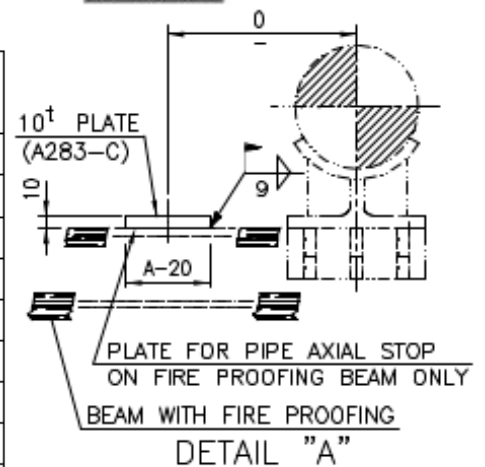
REV.
1

BY		CHK.	
DATE		APPR.	



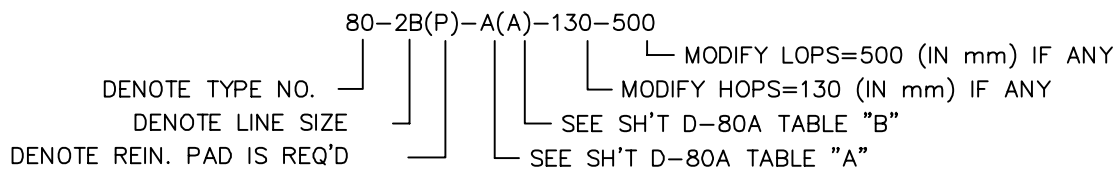
DIMENSIONS (mm)

LINE SIZE	A	B	C	D	E
1/2"	100	100	CUT FROM H200X100X5.5X8	-	-
3/4"	100	100		-	-
1"	100	100		-	-
1 1/2"	100	100		-	-
2"	100	100		-	-
2 1/2"	100	100		-	-
3"	100	100		-	-
4"	100	100	CUT FROM H200X200X8X12	-	-
5"	100	100		-	-
6"	200	100		70	-
8"	200	100		70	-
10"	200	200		70	-
12"	200	200		70	-
14"	200	200	FAB. FROM 12 ^t PLATE	70	-
16"	320	200		100	12
18"	320	200		100	12
20"	320	200		100	12
24"	370	200		100	12



NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:



2. LENGTH OF PIPE SHOE (SAY LOPS) TO SUIT BEAM WIDTH IN FIELD.

3. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



中鼎工程股份有限公司

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DETAIL OF PIPE SUPPORT

TYPE-80

(1 OF 2)

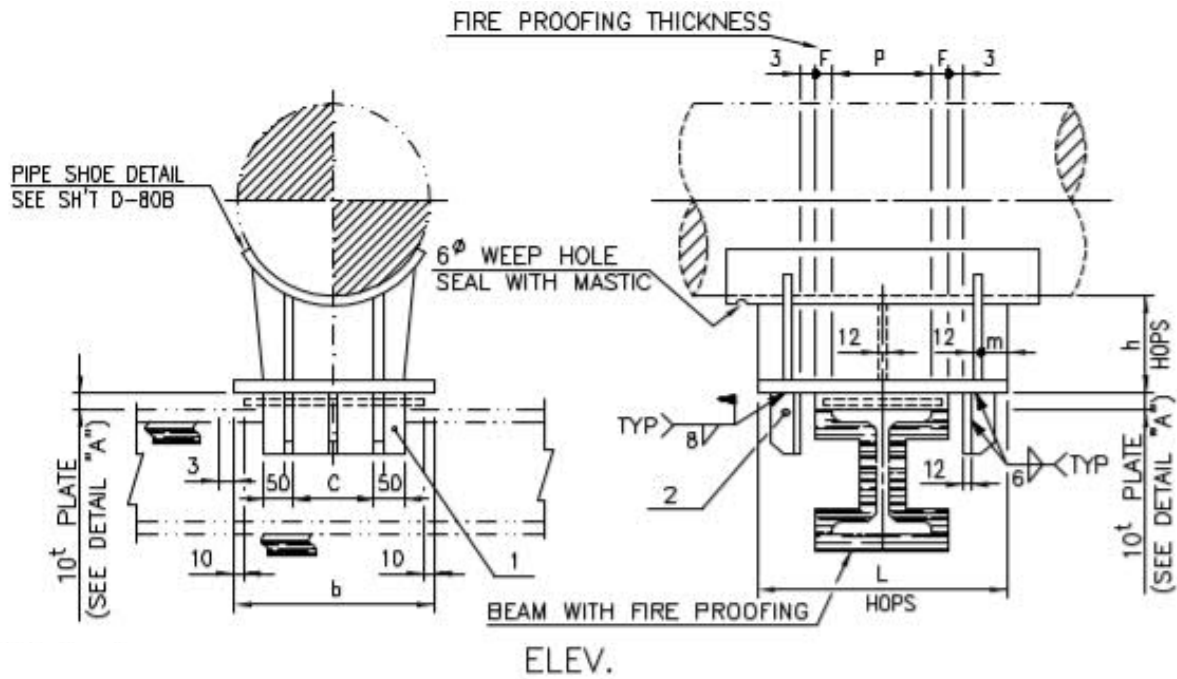
ENGINEERING STANDARD

D7TS-701-E

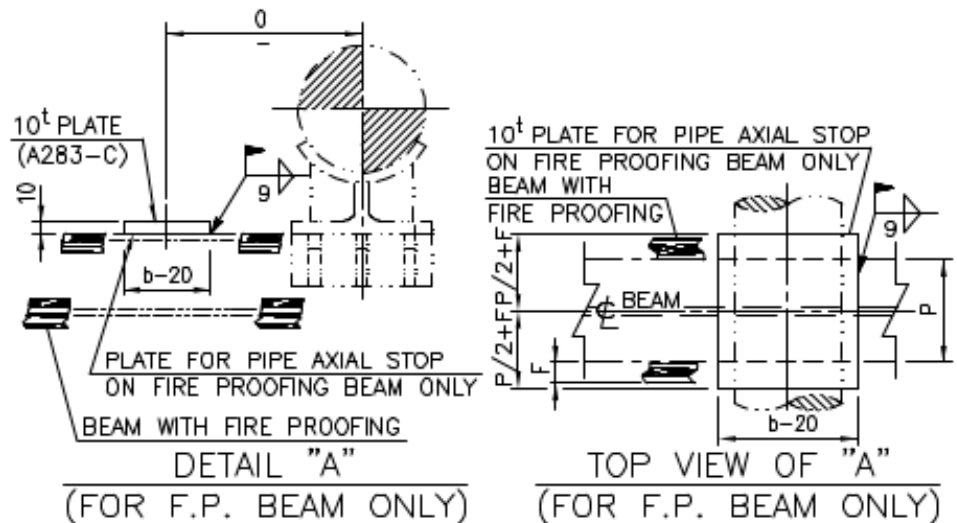
D-95

REV.
1BY
DATECHK.
APPR.

FOR LINE SIZE 26"-42"



NO.	1	2	
PIECE	2	6	
PIPE SIZE	PLATE	PLATE	C
26"	12 THK	12 THK	230
28"			240
30"			250
32"			260
36"			300
40"			335
42"			350



1. FOR DIMENSIONS AND INFORMATION NOT SHOWN ON THIS DWG. SEE SH'T D-80B.

2. DESIGNATION NUMBER DENOTE AS FOLLOW :

80-30B-A(A)-130-500

DENOTE TYPE NO. | DENOTE LINE SIZE | SEE NOTE 3 | MODIFY LOPS=500 (IN mm) IF ANY
MODIFY HOPS=130 (IN mm) IF ANY
SEE SH'T D-80A TABLE "B"

3. TYPE-A FOR INSULATION THICKNESS ≤ 100

TYPE-B FOR INSULATION THICKNESS $> 100 \leq 150$

TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED

4. LOADING CONDITION : CARBON STEEL SADDLE WITH MAX. LINE TEMPERATURE 371°C

5. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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CTCI CORPORATION

DETAIL OF PIPE SUPPORT

TYPE-80

(2 OF 2)

ENGINEERING STANDARD

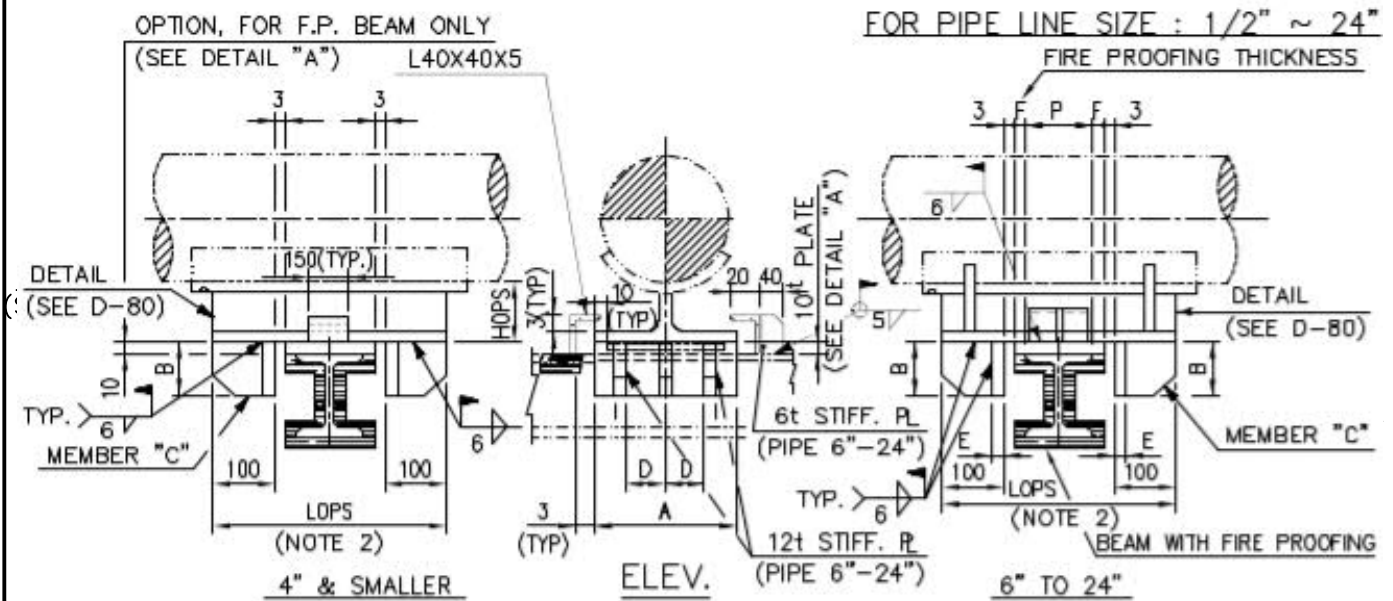
D7TS-701-E

D-96

REV.
1

BY
DATE

CHK.
APPR.



DIMENSIONS (mm)

LINE SIZE	A	B	C	D	E
1/2"	100	100	CUT FROM H200X100X5.5X8	-	-
3/4"	100	100		-	-
1"	100	100		-	-
1 1/2"	100	100		-	-
2"	100	100		-	-
2 1/2"	100	100		-	-
3"	100	100		-	-
4"	100	100		-	-
5"	100	100	CUT FORM H200X200X8X12	-	-
6"	200	100		70	-
8"	200	100		70	-
10"	200	200		70	-
12"	200	200		70	-
14"	200	200		70	-
16"	320	200	FAB. FORM 12 ^t PLATE	100	12
18"	320	200		100	12
20"	320	200		100	12
24"	370	200		100	12

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

83-2B(P)-A(A)-130-500

DENOTE TYPE NO.

DENOTE LINE SIZE

DENOTE REIN. PAD IS REQ'D

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

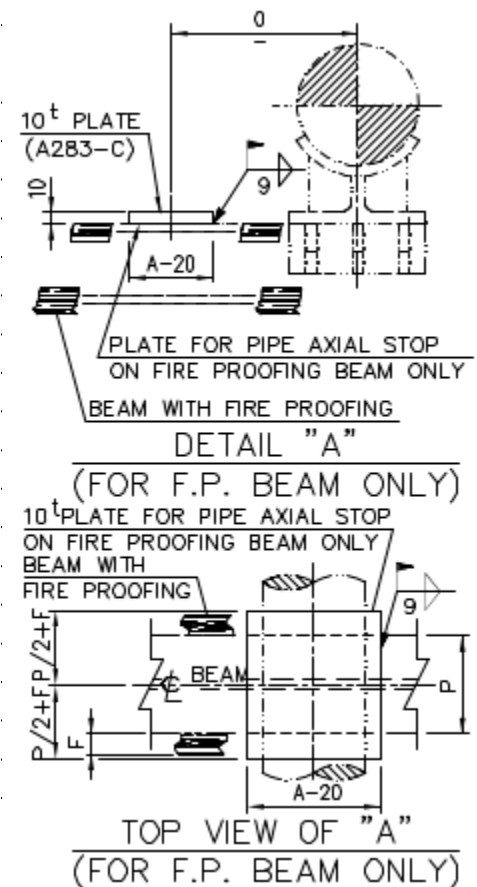
SEE SH'T D-80A TABLE "B"

SEE SH'T D-80A TABLE "A"

2. THE MAX. LENGTH OF THIS MEMBER SHALL NOT EXCEED THE WIDTH OF BEAM.

3. PIPE SHOE LENGTH (SAY LOPS) TO SUIT BEAM WIDTH IN FIELD.

4. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-83

(1 OF 2)

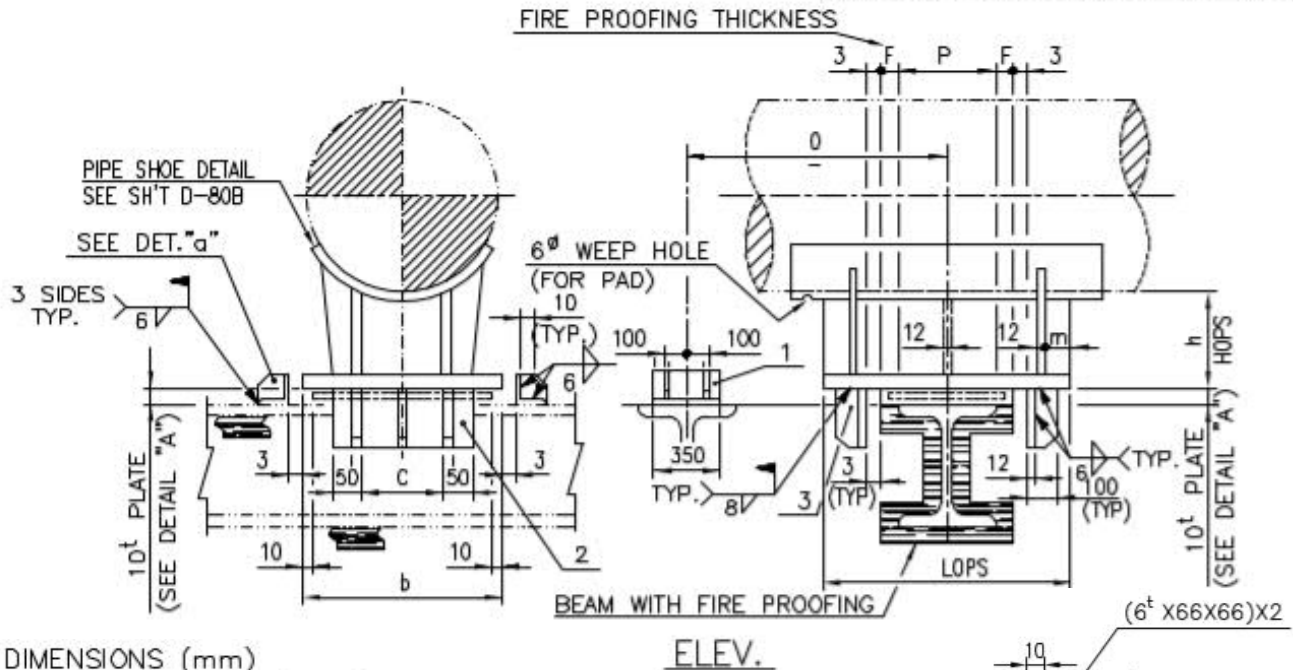
ENGINEERING STANDARD

D7TS-701-E

D-101

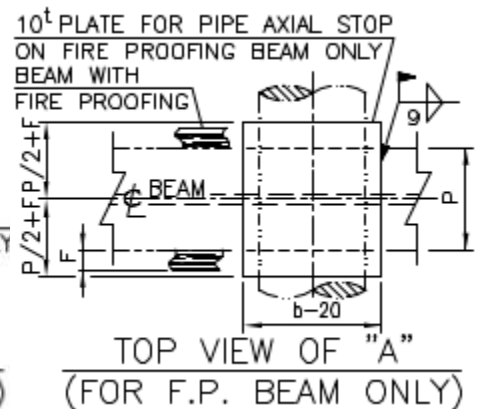
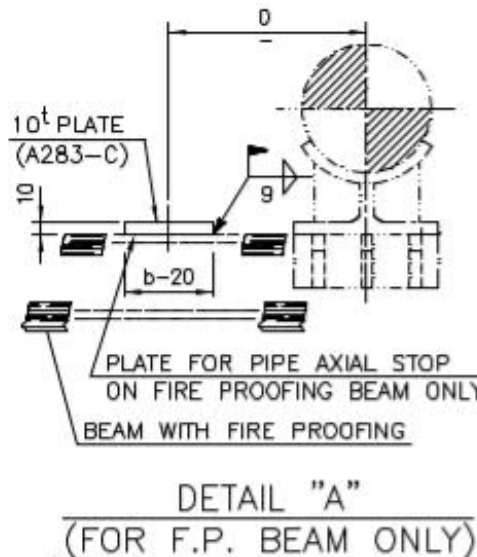
REV.
1BY
DATECHK.
APPR.

FOR PIPE LINE SIZE : 26" ~ 42"



DIMENSIONS (mm)

NO.	1	2	3	
PIECE	2	2	6	
PIPE SIZE	GUIDE	PLATE	PLATE	C
26"	L75X75X9	12 TH'K	12 TH'K	230
28"				240
30"				250
32"				260
36"				300
40"				335
42"				350



NOTES :

- FOR DIMENSIONS AND INFORMATION NOT SHOWN ON THIS DWG. SEE SH'T D-80B.
- DESIGNATION NUMBER DENOTE AS FOLLOWS :
 83-30B-A(A)-130-500
 DENOTE TYPE NO. MODIFY LOPS=500 (IN mm) IF ANY
 DENOTE LINE SIZE MODIFY HOPS=130 (IN mm) IF ANY
SEE SH'T D-80A TABLE "B"
SEE NOTE 3
- TYPE-A FOR INSULATION THICKNESS < 100
 TYPE-B FOR INSULATION THICKNESS > 100 ≤ 150
 TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED
- LOADING CONDITION : CARBON STEEL SADDLE WITH MAX. LINE TEMPERATURE 371°C
- HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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CTCI CORPORATION

DETAIL OF PIPE SUPPORT

TYPE-83

(2 OF 2)

ENGINEERING STANDARD

D7TS-701-E

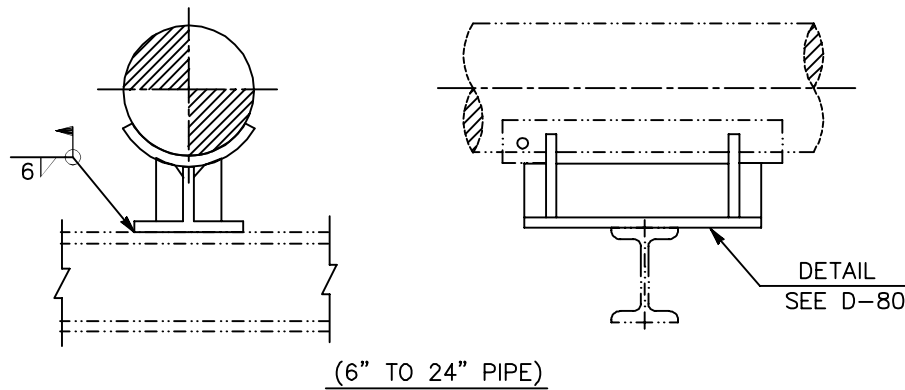
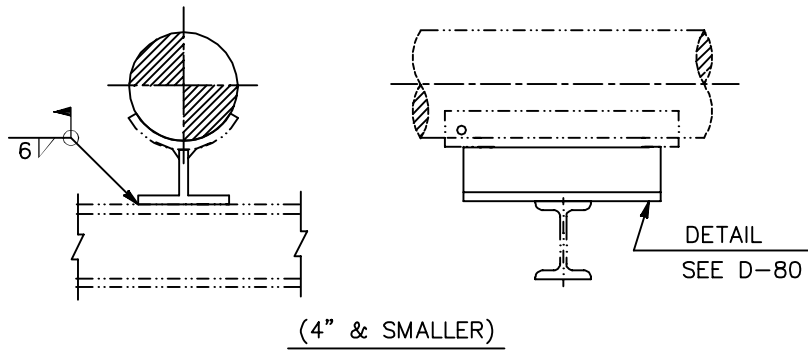
D-102

REV.
1

BY
DATE

CHK.
APPR.

FOR LINE SIZE 1/2"-24"



NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:

85-2B(P)-A(A)-130-500

DENOTE TYPE NO. ————

DENOTE LINE SIZE ————

DENOTE REIN. PAD IS REQ'D ————

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "B"

SEE SH'T D-80A TABLE "A"

2. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE



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DETAIL OF PIPE SUPPORT
TYPE-85

ENGINEERING STANDARD

D7TS-701-E

BY			CHK.		
DATE			APPR.		

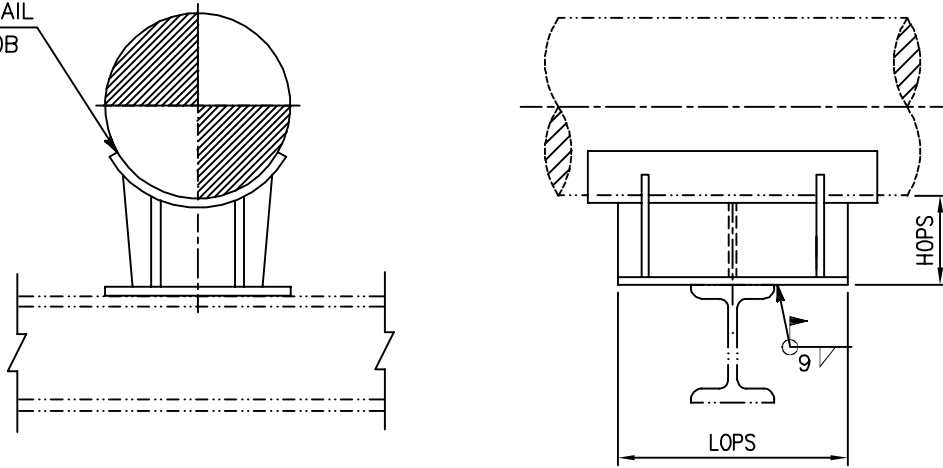
(1 OF 2)

D-105

REV.
1

FOR LINE SIZE 26"-42"

PIPE SHOE DETAIL
SEE SH'T D-80B



ELEV.

NOTES :

1. FOR DIMENSIONS AND INFORMATION NOT SHOWN ON THIS DWG. SEE SH'T D-80B.

2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

85-30B-A(A)-130-500

DENOTE TYPE NO.		85		30B		A(A)		130		500
DENOTE LINE SIZE										

MODIFY LOPS=500 (IN mm) IF ANY
 MODIFY HOPS=130 (IN mm) IF ANY
 SEE SH'T D-80A TABLE "B"
 SEE NOTE 3

3. TYPE-A FOR INSULATION THICKNESS ≤ 100
 TYPE-B FOR INSULATION THICKNESS > 100 AND ≤ 150
 TYPE-C FOR INSULATION THICKNESS > 150 OR SPECIAL USED.

4. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT

TYPE-85

(2 OF 2)

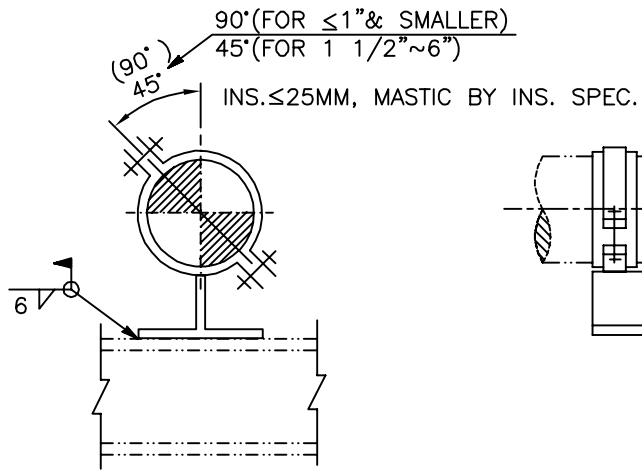
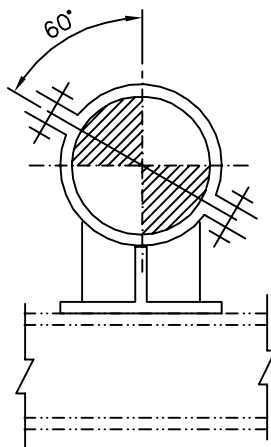
ENGINEERING STANDARD

D7TS-701-E

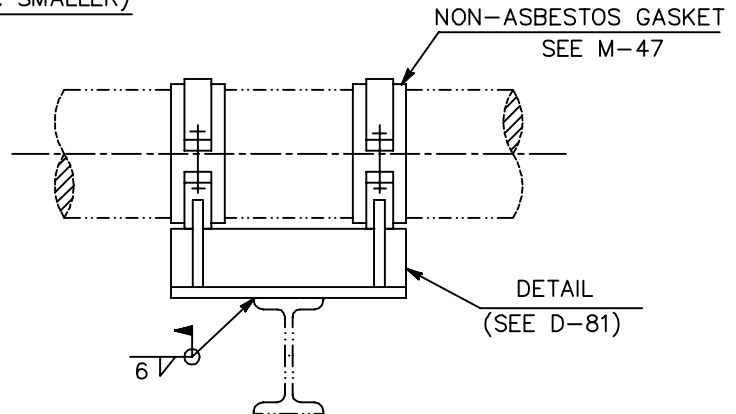
D-106

REV.
1

BY			CHK.		
DATE			APPR.		

FOR LINE SIZE $1/2"-24"$ NON-ASBESTOS GASKET
SEE M-47 (SEE NOTE 3)DETAIL
(SEE D-81)

(4" & SMALLER)



(6" TO 24" PIPE)

ELEV.

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS:

86-2B-A(A)-130-500

DENOTE TYPE NO. ————

DENOTE LINE SIZE ————

SEE SH'T D-80A TABLE "A" ————

MODIFY LOPS=500 (IN mm) IF ANY

MODIFY HOPS=130 (IN mm) IF ANY

SEE SH'T D-80A TABLE "B"

2. THIS TYPE IS CLAMP ON PIPE SHOE.(MAX. LINE TEMP.: 371°C)
3. NON-ASBESTOS GASKET IS USED FOR ALLOY STEEL AND STAINLESS STEEL LINES.
4. HOPS=HEIGHT OF PIPE SHOE, LOPS=LENGTH OF PIPE SHOE.



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DETAIL OF PIPE SUPPORT
TYPE-86

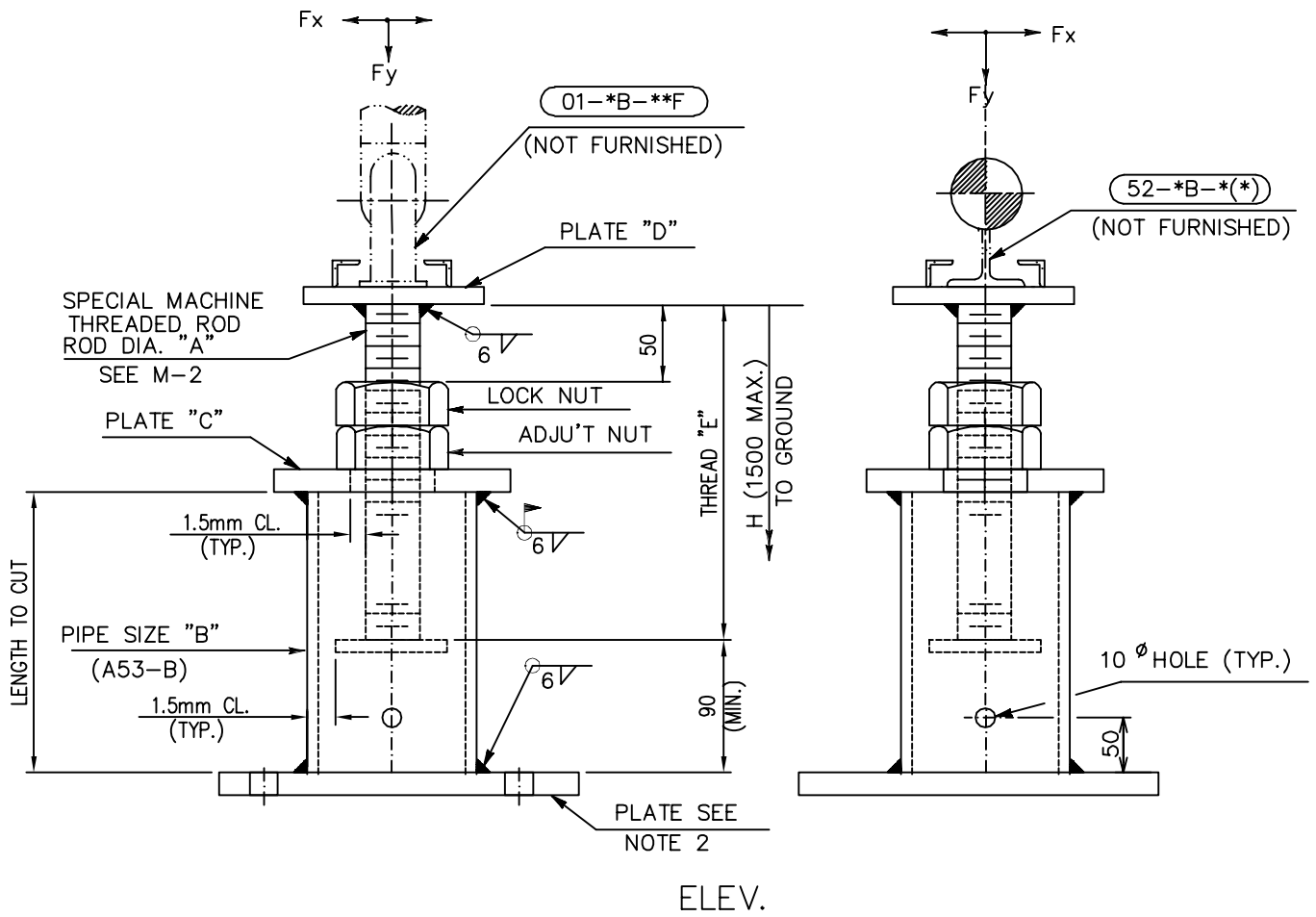
ENGINEERING STANDARD

D7TS-701-E

D-107

REV.
1

BY		CHK.	
DATE		APPR.	



DIMENSIONS (mm)

A	B	C	D	E
1"	3"STD	120X120X9 ^t	210X210X9 ^t	190
1 1/2"	4"STD	150X150X9 ^t	290X290X9 ^t	215
2"	6"STD	200X200X12 ^t	290X290X12 ^t	245
2 1/2"	8"STD	250X250X16 ^t	390X390X16 ^t	275

NOTES :

1. DIMENSION "H" SHALL BE CUT TO SUIT IN FIELD.
2. THIS TYPE SHALL BE USED WITH M-42 TO DETERMINE THE LOWER COMPONENT TYPE. IN THIS DETAIL USE TYPE-G, T & R ONLY.
3. DESIGNATION NUMBER DENOTE AS FOLLOWS :

87-1-10G
 DENOTE TYPE NO. DENOTE THE LOWER COMPONENT TYPE SEE M-42
 DENOTE ROD SIZE "A" DENOTE DIMENSION "H" (IN 100mm)

4. IF THE LOWER COMPONENT TYPE-G IS USED, DIMENSION "H" SHALL BE COUNTED FROM THE ELEVATION OF THE LOWEST POINT OF PAVING.



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DETAIL OF PIPE SUPPORT
TYPE-87

ENGINEERING STANDARD

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REV.
1

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DATE

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APPR.

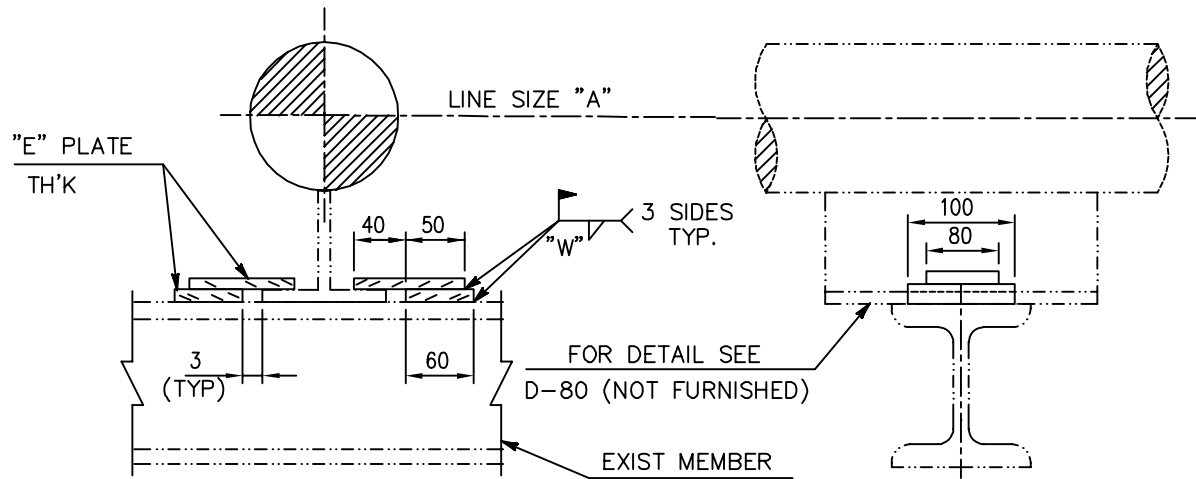


FIG. A

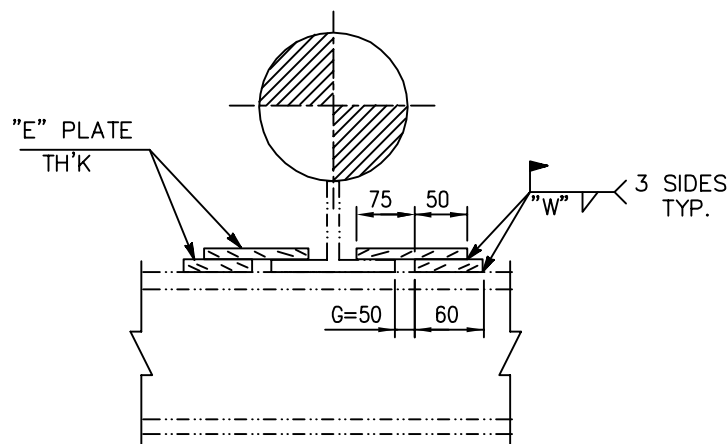


FIG. B

DIMENSIONS (mm)

LINE SIZE A	3/4"	1"	1 1/2"	2"	3"	4"	6"	8"	10"	12"	14"	16"	18"	20"	24"
E	9	9	9	9	9	9	12	12	12	12	12	12	12	12	12
W	6	6	6	6	6	6	10	10	10	10	10	10	10	10	10

NOTES :

1. DESIGNATION NUMBER DENOTE AS FOLLOWS :

102-1B-A
 ——— DENOTE FIG. NO.
 ——— DENOTE LINE SIZE
 ——— DENOTE TYPE NO.



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DETAIL OF PIPE SUPPORT
TYPE-102

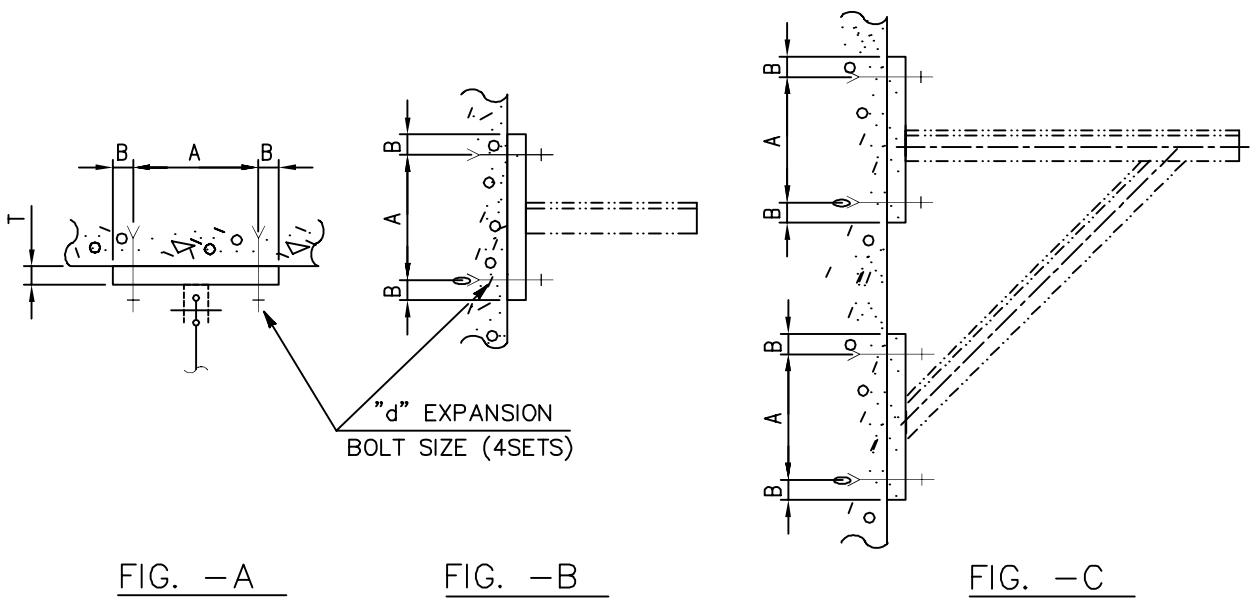
ENGINEERING STANDARD

D7TS-701-E

D-111

REV.
1

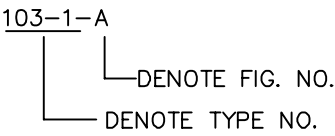
BY		CHK.	
DATE		APPR.	



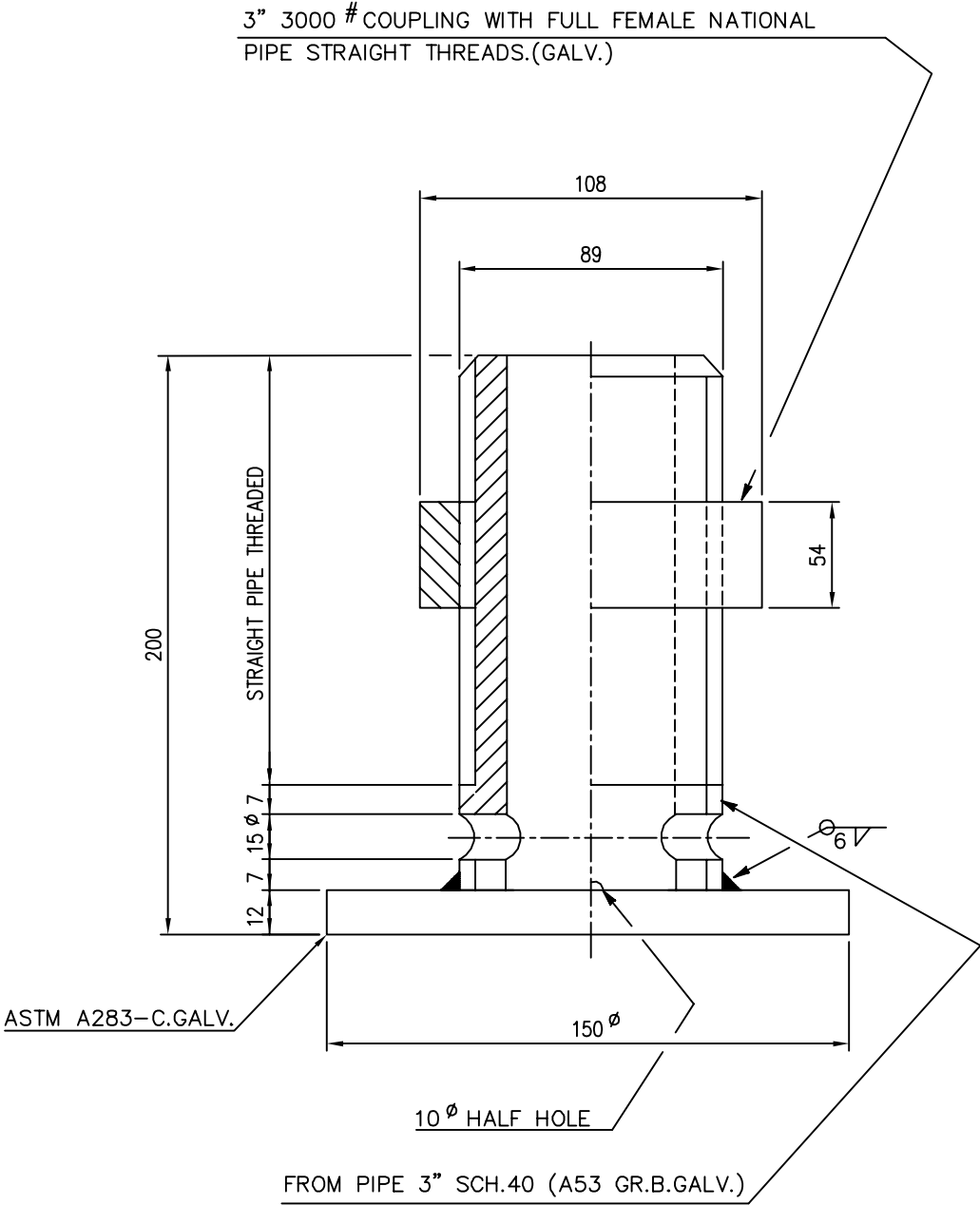
SUPPORT NO.	A (mm)	B (mm)	T (mm)	d
103-1	150	15	9	1/4"X4
103-2	150	20	9	3/8"X4
103-3	200	25	16	1/2"X4
103-4	250	30	19	5/8"X4
103-5	250	35	19	3/4"X4
103-6	250	40	19	7/8"X4

NOTES:

1. DESIGNATION NUMBER DENOTE AS FOLLOW:



 中鼎工程股份有限公司 CTCI CORPORATION						ENGINEERING STANDARD	
DETAIL OF PIPE SUPPORT TYPE-103						D7TS-701-E	
						D-112	REV. 1
BY			CHK.				
DATE			APPR.				



中鼎工程股份有限公司
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SPECIAL BASE PLATE

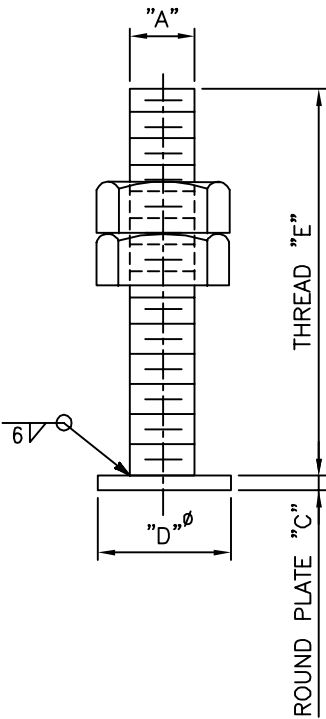
ENGINEERING STANDARD

D7TS-701-E

BY			CHK.		
DATE			APPR.		

M-1

REV.
1



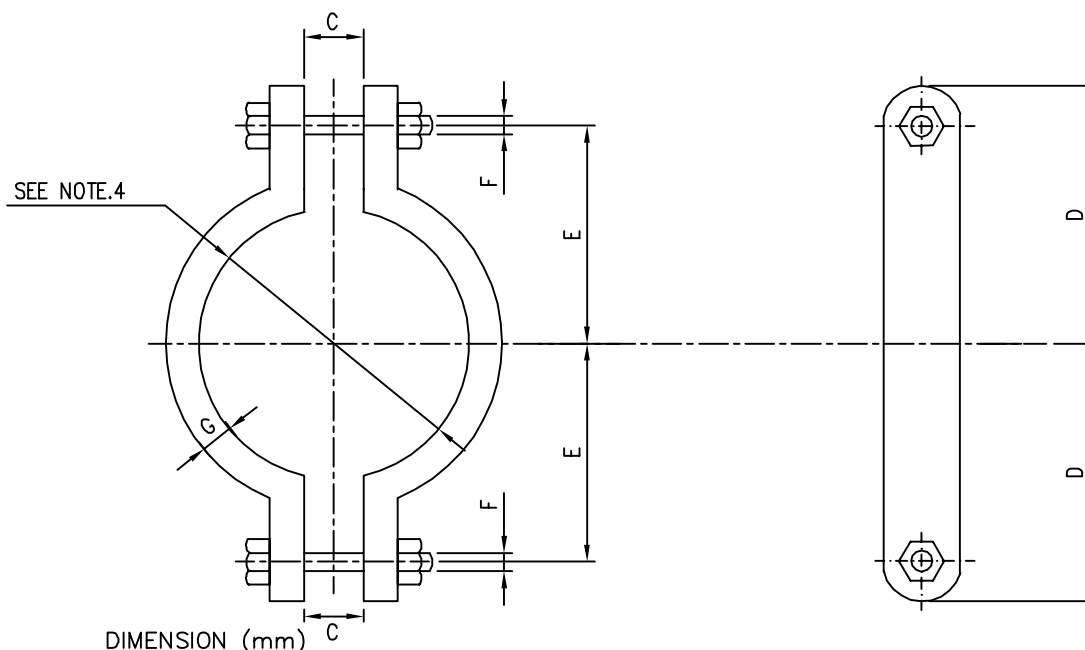
DIMENSIONS (mm)

TYPE	A	E	C ^t	D [∅]
SMTR-1	1"	190	9	72
SMTR-1 1/2	1 1/2"	215	9	96
SMTR-2	2"	245	12	148
SMTR-2 1/2	2 1/2"	275	12	197

- NOTES :
- 1. MATERIAL : SPECIAL MACHINE FULL THREADED ROD AND TWO HEX. NUTS
ASTM A283-C GALV./A307-B GALV./A563-A GALV.
 - 2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

SMTR - 1
└──┬── DENOTE DIM. "A"
 └── DENOTE SPECIAL MACHINE THREADED ROD

 中鼎工程股份有限公司 CTCI CORPORATION						ENGINEERING STANDARD	
SPECIAL MACHINE THREADED ROD						D7TS-701-E	
						M-2	REV. 1
BY			CHK.				
DATE			APPR.				



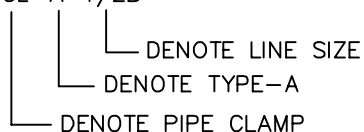
DIMENSION (mm)

TYPE	LINE SIZE	MAX. RECOM. LOAD (Kg)		C	D	E	F	G
		340°C	371°C					
PCL-A-1/2B	1/2"	220	200	11	39	25	5/16"	3 X 25
PCL-A-3/4B	3/4"	220	200	11	42	29	5/16"	3 X 25
PCL-A- 1 B	1"	220	200	11	48	35	5/16"	3 X 25
PCL-A-1 1/4B	1 1/4"	220	200	11	50	37	5/16"	3 X 25
PCL-A-1 1/2B	1 1/2"	365	325	13	55	41	5/16"	3 X 25
PCL-A- 2 B	2"	475	420	13	70	54	1/2"	6 X 25
PCL-A-2 1/2B	2 1/2"	475	420	16	83	67	1/2"	6 X 25
PCL-A- 3 B	3"	475	420	16	90	75	1/2"	6 X 25
PCL-A-3 1/2B	3 1/2"	475	420	16	97	81	1/2"	6 X 25
PCL-A- 4 B	4"	475	420	19	111	92	5/8"	6 X 32
PCL-A- 5 B	5"	475	420	19	125	106	5/8"	6 X 32
PCL-A- 6 B	6"	735	655	22	149	127	3/4"	9 X 38
PCL-A- 8 B	8"	735	655	25	178	156	3/4"	9 X 38
PCL-A- 10 B	10"	1130	1010	25	217	189	7/8"	12 X 51
PCL-A- 12 B	12"	1130	1010	25	243	214	7/8"	12 X 51
PCL-A- 14 B	14"	1130	1010	29	270	235	7/8"	12 X 64
PCL-A- 16 B	16"	1130	1010	29	295	260	7/8"	12 X 64
PCL-A- 18 B	18"	1390	1240	32	330	295	1"	16 X 64
PCL-A- 20 B	20"	1390	1240	35	359	324	1 1/8"	16 X 64
PCL-A- 24 B	24"	1390	1240	41	429	387	1 1/4"	16 X 76

NOTES :

1. MATERIAL: CARBON STEEL GALV.
2. DESIGNATION NUMBER DENOTE AS FOLLOWS :

PCL-A-1/2B



3. REF. ANVIL FIG. 212. OR EQ.
4. CLAMP I.D. = LINE O.D. + 3mm



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PIPE CLAMP
TYPE-A

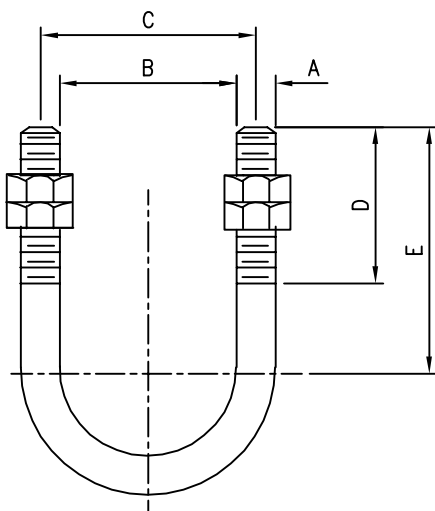
ENGINEERING STANDARD

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M-4

REV.
1

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DIMENSIONS (mm)

TYPE	LINE SIZE	ROD SIZE A	MAX. RECOM. LOAD (Kg)		B	C	D	E
			340°C	371°C				
UB - 1/4 B	1/4"	1/4"	220	220	18	24	36	39
UB - 1/2 B	1/2"	1/4"	220	200	24	30	40	43
UB - 3/4 B	3/4"	1/4"	220	200	29	35	40	45
UB -1B	1"	1/4"	220	200	35	41	40	49
UB -1 1/4 B	1 1/4"	3/8"	555	495	43	52	58	65
UB -1 1/2 B	1 1/2"	3/8"	555	495	51	60	58	68
UB -2B	2"	3/8"	555	495	62	71	58	74
UB -2 1/2 B	2 1/2"	1/2"	1025	920	75	87	64	85
UB -3B	3"	1/2"	1025	920	91	103	64	93
UB -3 1/2 B	3 1/2"	1/2"	1025	920	103	116	64	99
UB -4B	4"	1/2"	1025	920	116	129	67	108
UB -5B	5"	1/2"	1025	920	143	156	67	122
UB -6B	6"	5/8"	1645	1470	171	187	79	143
UB -8B	8"	5/8"	1645	1470	222	238	79	169
UB -10B	10"	3/4"	2460	2155	276	295	102	213
UB -12B	12"	7/8"	3425	3060	327	349	111	245
UB -14B	14"	7/8"	3425	3060	359	381	111	261
UB -16B	16"	7/8"	3425	3060	410	432	111	286
UB -18B	18"	1"	4500	4015	460	486	117	316
UB -20B	20"	1"	4500	4015	511	537	121	345
UB -24B	24"	1"	4500	4015	613	638	121	396
UB -30B	30"	1"	4500	4015	765	791	121	472

NOTES :

- MATERIAL : CARBON STEEL GALVANIZED U-BOLT AND FOUR FINISHED HEX. NUTS.
(FOR CARBON STEEL PIPE)
STAINLESS STEEL U-BOLT AND FOUR FINISHED HEX. NUTS.
(FOR STAINLESS STEEL PIPE)

- DESIGNATION NUMBER DENOTE AS FOLLOWS :

UB-1/2B

_____ DENOTE LINE SIZE
_____ DENOTE CARBON STEEL U-BOLT

UBS-1/2B

_____ DENOTE LINE SIZE
_____ DENOTE STAINLESS STEEL U-BOLT



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U-BOLT

ENGINEERING STANDARD

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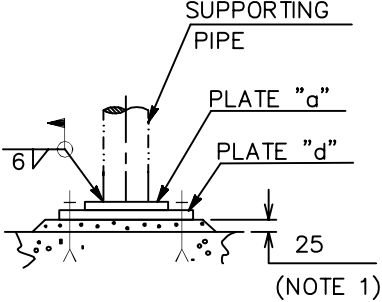
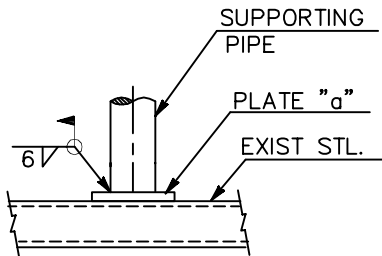
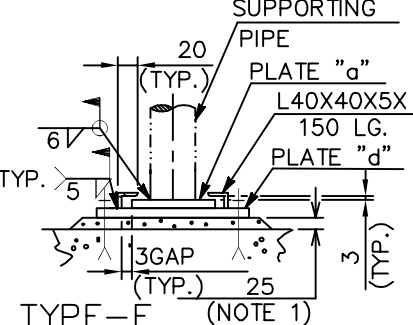
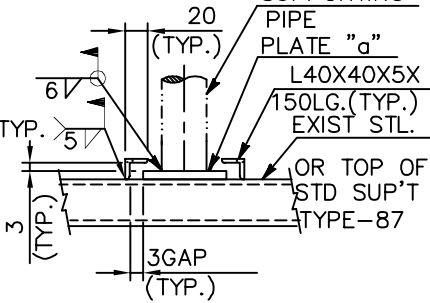
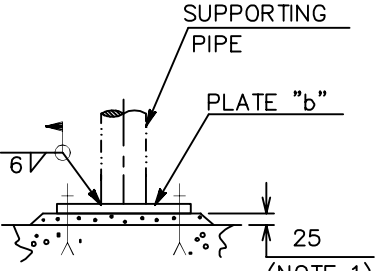
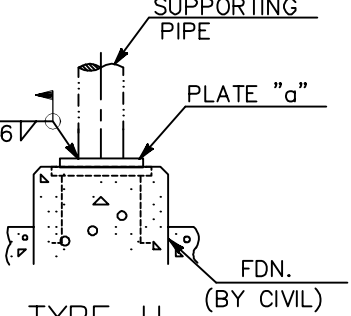
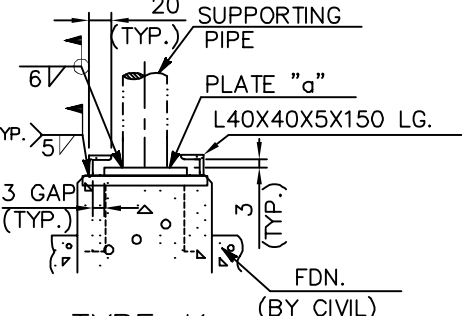
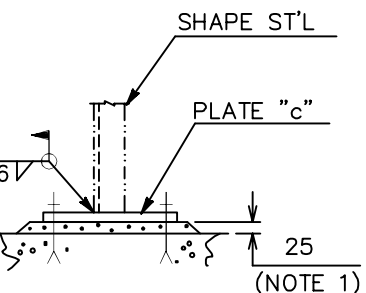
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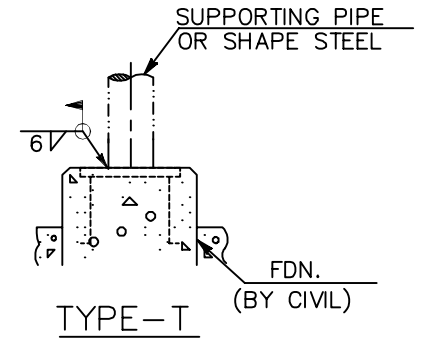
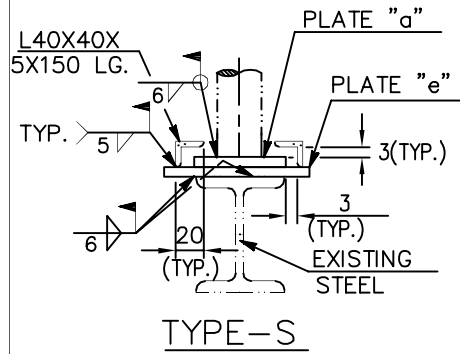
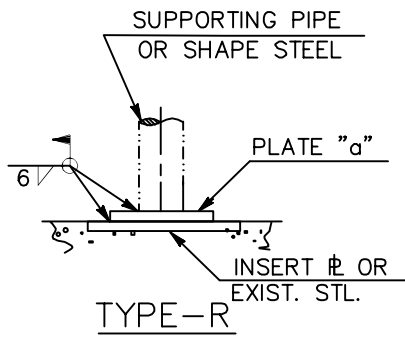
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1

	 TYPE-B	 TYPE-C
	 TYPE-E	 TYPE-F
 TYPE-G	 TYPE-H	
 TYPE-K	 TYPE-L	

NOTE:1. THE MATERIAL OF GROUT FOLLOWS BY CIVIL SPECIFICATION.
"XF11A-0000-001 DESIGN SPECIFICATION FOR FOUNDATION & MISCELLANEOUS CIVIL WORK."

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LOWER COMPONENT OF
BASE SUPPORT

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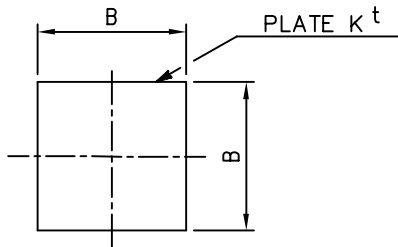


PLATE "a"

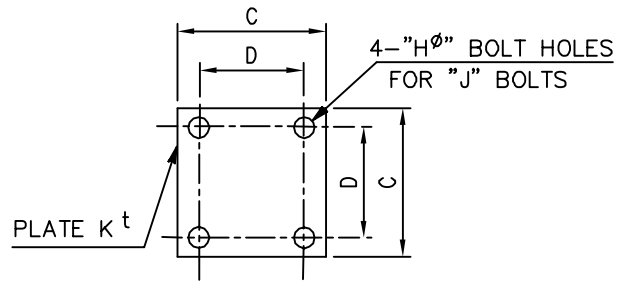


PLATE "b"

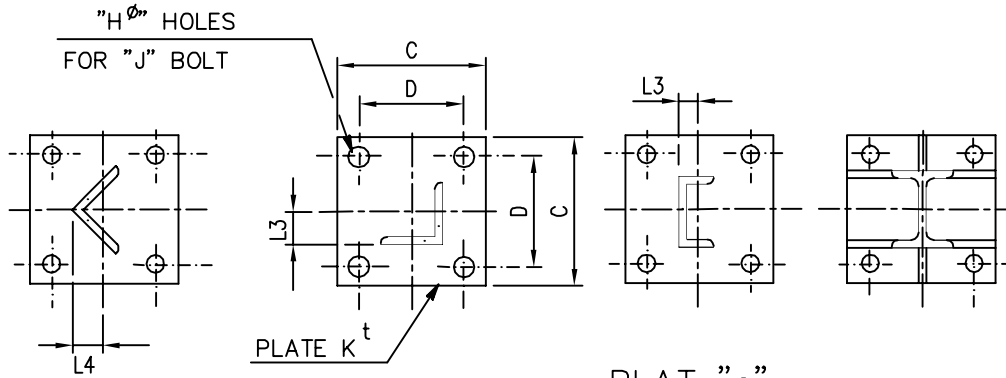


PLATE "c"

SUP'TING STEEL	L3	L4
L50	25	15
L75	35	25
L100	50	—
C125	30	—
C150	30	—

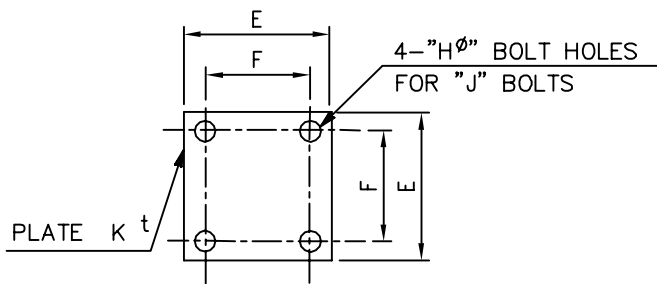


PLATE "d"

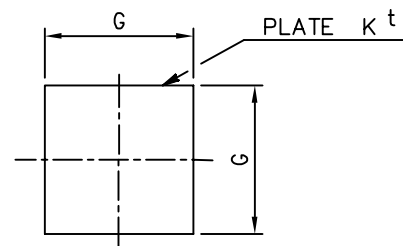


PLATE "e"

SUPPORTING PIPE	SUPPORTING STEEL	B	C	D	E	F	G	EXP. BOLT H	J	K
1"~3"	L50,L65	150	180	110	290	220	200	19	5/8"	9
4"~6"	L75,L100	230	260	190	370	300	280	19	5/8"	9
8"	C125	330	380	300	490	410	380	19	5/8"	16
10"	C150,H125	330	380	300	490	410	380	22	3/4"	16
12"	H150,H200	380	500	410	560	470	430	26	1"	16

NOTES :

1. IF TYPE-B,C, & H USED FOR SUPPORT NO. 09. AND 10, PLATE "d" SHALL BE DELETED.



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LOWER COMPONENT OF
BASE SUPPORT

(3 OF 3)

ENGINEERING STANDARD

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TYPE	D	L	THREAD LENGTH	SIZE OF R.C HOLE TO BE DRILLED	LOADING (NOTE 3)	
					TENSILE(Kg)	SHEAR(Kg)
EB – 1/4"	1/4"	76	19	6 Ø X32 L	693	892
EB – 3/8"	3/8"	89	32	10 Ø X38 L	1602	1905
EB – 1/2"	1/2"	114	35	12 Ø X57 L	2312	3300
EB – 5/8"	5/8"	127	38	16 Ø X70 L	3083	4503
EB – 3/4"	3/4"	152	38	20 Ø X83 L	4417	6322
EB – 1"	1"	178	57	24 Ø X95 L	5629	8400

EB - 1/4"
 └──┬── DENOTE DIMENSION "D"
 └──┴── DENOTE EXP. BOLT

3. THESE LOADINGS ARE BASED ON R.C. STRENGTH 210 KG/CM^2



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EXPANSION BOLT

D7TS-701-E

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